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-

How will ML know what to do if we provide no instruction?



“The dog has been in the habit of carrying this stick behind his master. Being a heavy stick the dog has held it tightly by the middle, and the marks of his teeth are very plainly visible. The dog's jaw, as shown in the space between these marks, is too broad in my opinion for a terrier and not broad enough for a mastiff. It may have been – yes, by Jove it is a curly-haired spaniel.”

Features should be correlated with the target label

In machine learning and pattern recognition, a feature is an individual measurable property or characteristic of a phenomenon.

Choosing **informative**, **discriminating** and **independent** features is a crucial element of effective algorithms in pattern recognition, classification and regression.

Features should be correlated with the target label



Shoe size (inches)

Shoe size (cm)

Age

Favorite food

Race

Shirt Color

Gender

Zip Code

Asleep/Awake

Genome sequence

Occupation

Eye color

Weight

Not all features are equally important



Shoe size (inches)

Age

Gender

Weight

Race

Genome sequence (highly complex)

Occupation (weak?)

Eye color (weak?)

Zip Code (weak?)

Shoe size (cm) (not independent)

Favorite food (no correlation)

Shirt Color (no correlation)

Asleep/Awake (no correlation)

Featurization is not the same as feature engineering

Featurization is a way to change some form of data (text data, graph data, time-series data,...) into a numerical vector.

Featurization is different from feature engineering. Feature engineering is just transforming the numerical features somehow so that the machine learning models work well. In feature engineering, features are already in the numerical form, whereas in featurization data not need to be in the form of numerical vector.

What sort of materials features do we have to pick from?

Electronic



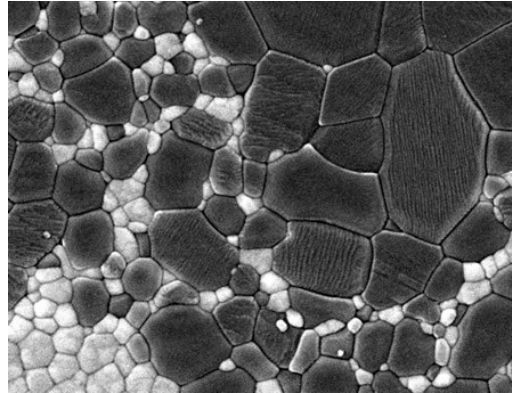
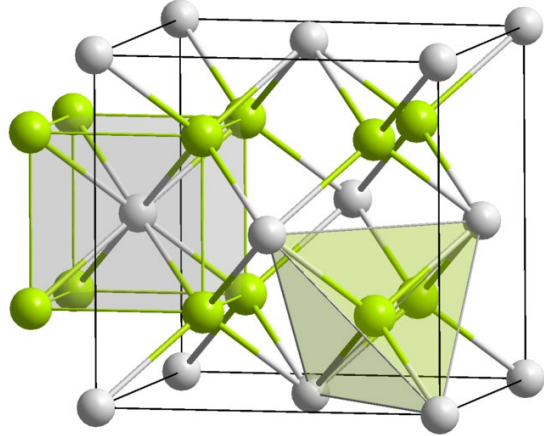
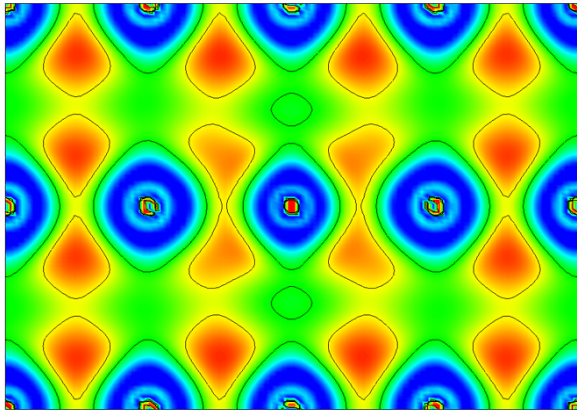
Atomic



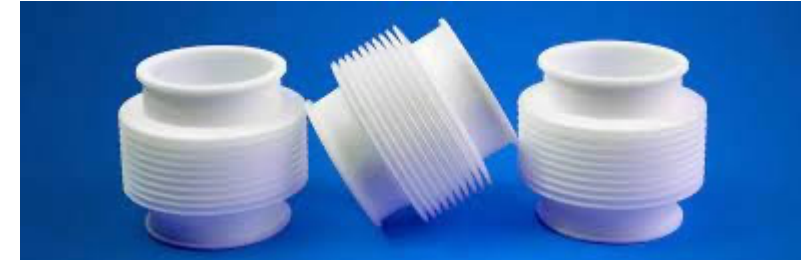
Microstructure



Macroscale



How do we teach machine learning chemistry?



How do we teach machine learning chemistry?



ceramic, hard, high
melting point, strong



metal, alloy, intermediate
melting point, ductile,
corrosion resistant

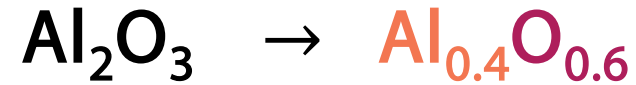


polymer, non-stick, low
melting point, low friction

We can use elemental data to create a composition-based feature vector

Element	Number	Atomic_Weight	Period	Group	Families	Metal	Nonmetal	Metalliod	Atomic_Radius
H	1	1.00794	1	1	7	0	1	0	0.53
He	2	4.002602	1	18	9	0	1	0	0.31
Li	3	6.941	2	1	1	1	0	0	1.67
Be	4	9.01218	2	2	2	1	0	0	1.12
B	5	10.811	2	13	6	0	0	1	0.87
C	6	12.011	2	14	7	0	1	0	0.67
N	7	14.00674	2	15	7	0	1	0	0.56
O	8	15.9994	2	16	7	0	1	0	0.48
F	9	18.998403	2	17	8	0	1	0	0.42
Ne	10	20.1797	2	18	9	0	1	0	0.38
Na	11	22.989768	3	1	1	1	0	0	1.9
Mg	12	24.305	3	2	2	1	0	0	1.45
Al	13	26.981539	3	13	5	1	0	0	1.18
Si	14	28.0855	3	14	6	0	1	0	1.11

Unique formulae get unique vectors

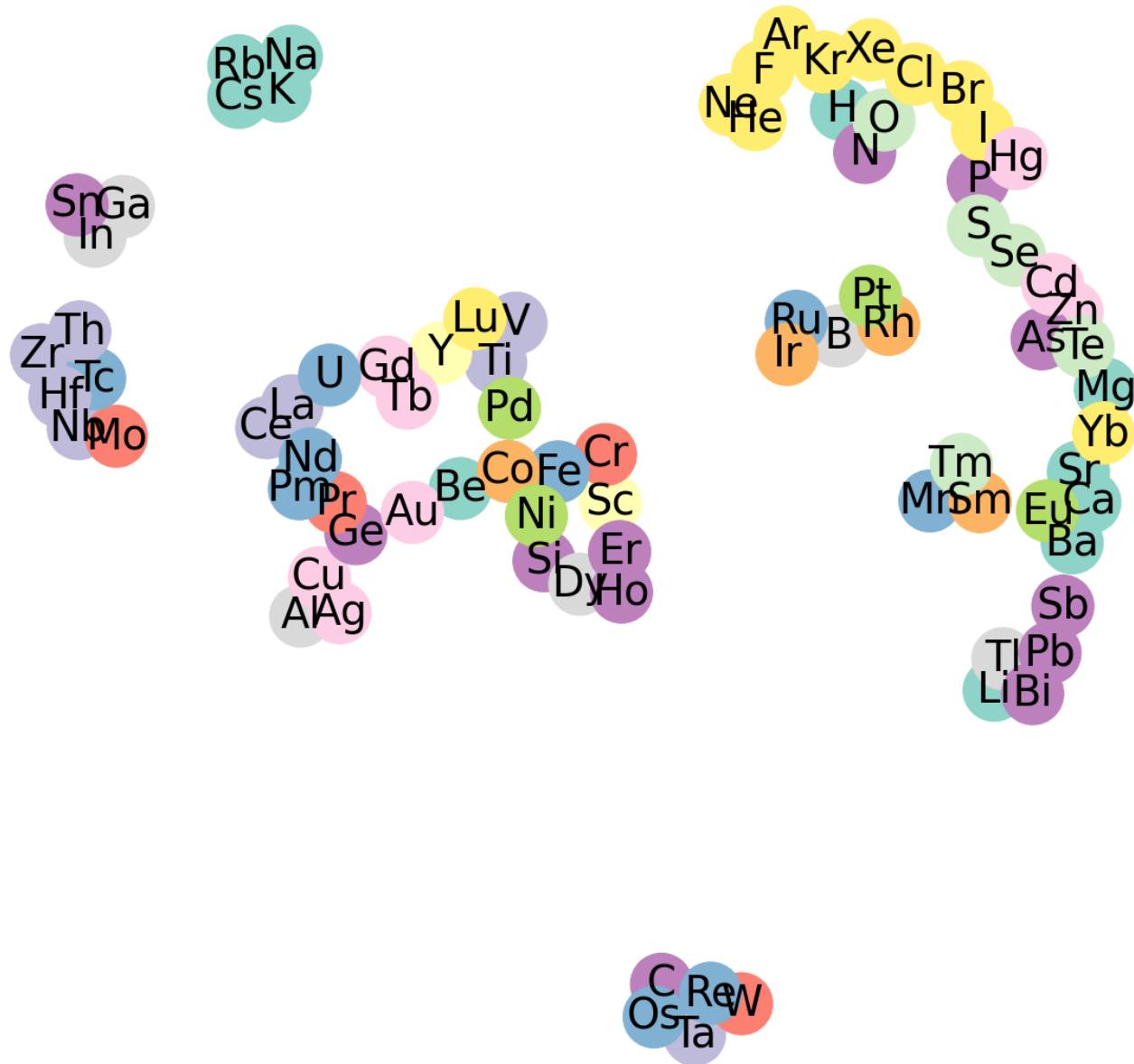


C	6	12.011	2	14	7	0	1	0	0.67
N	7	14.00674	2	15	7	0	1	0	0.56
O	8	15.9994	2	16	7	0	1	0	0.48
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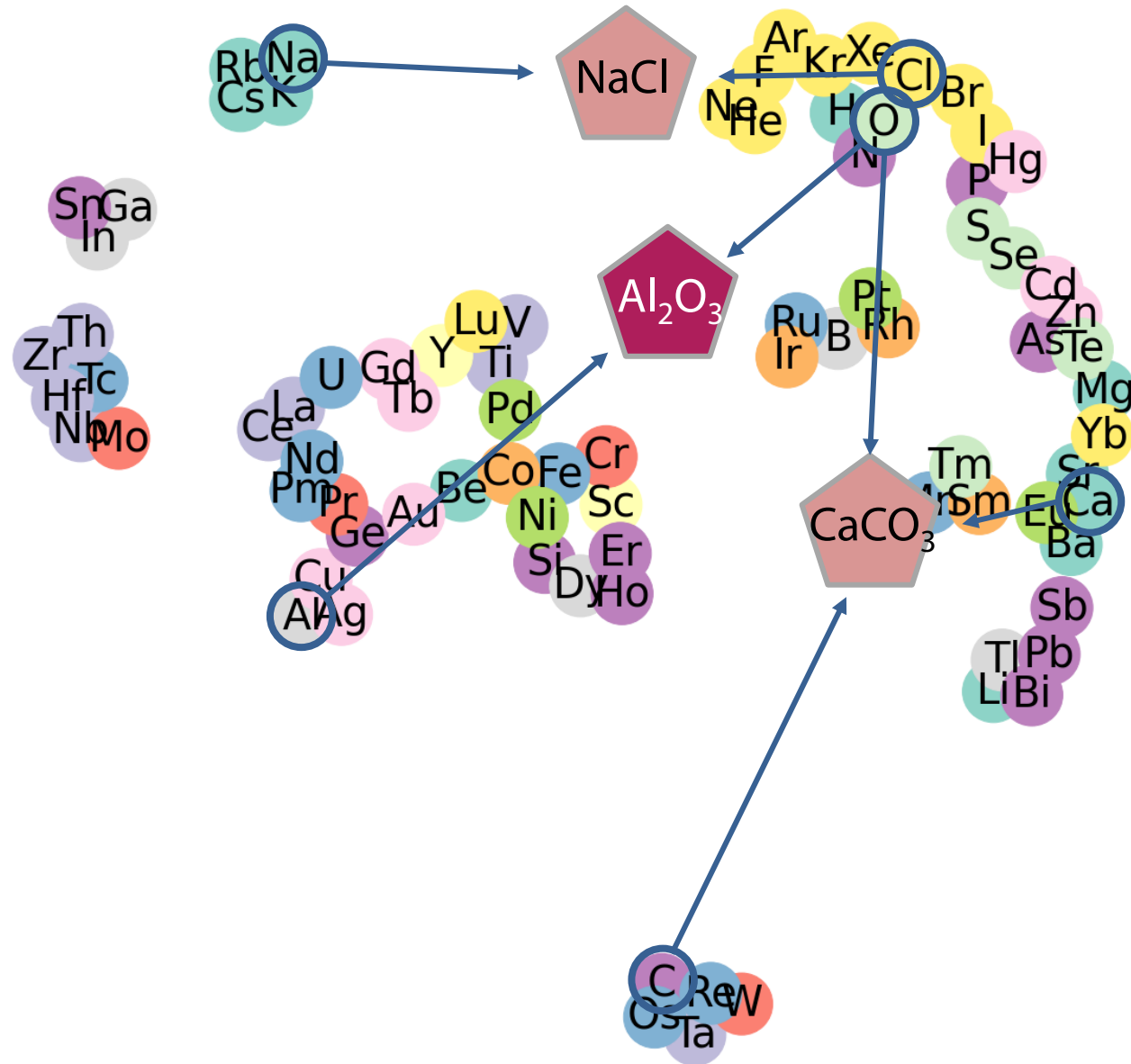
$$\text{Al}_{0.4}\text{O}_{0.6} = [8 \times 0.6 + 13 \times 0.4, \\ 16 \times 0.6 + 27 \times 0.4, \\ \dots, \\ 0.48 \times 0.6 + 1.18 \times 0.4]$$

$$\text{Al}_{0.4}\text{O}_{0.6} = [10, \\ 20.4, \\ \dots, \\ 3.35]$$

Projections of this CBFV group similar elements together!

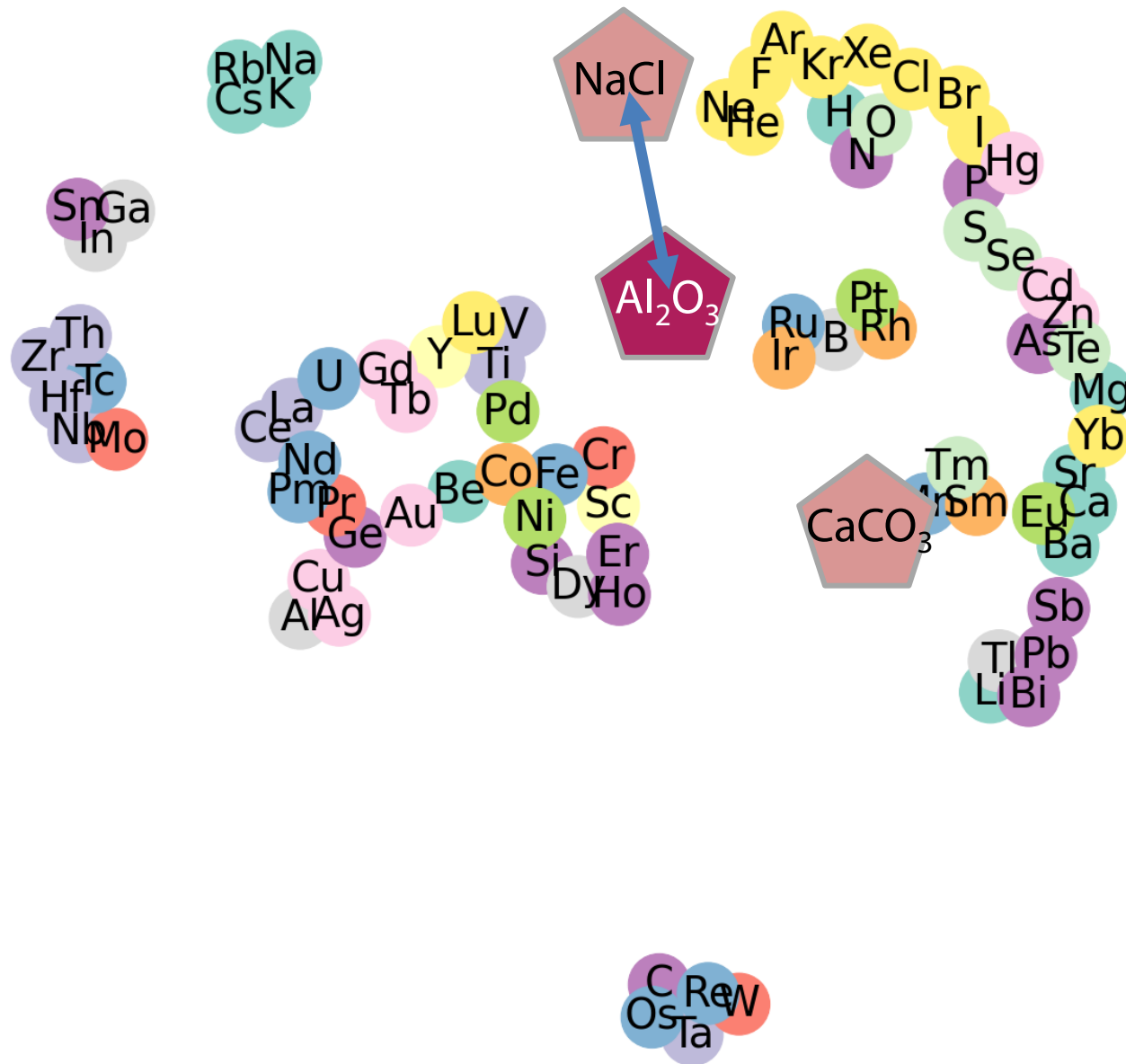


Projections of this CBFV group similar elements together!



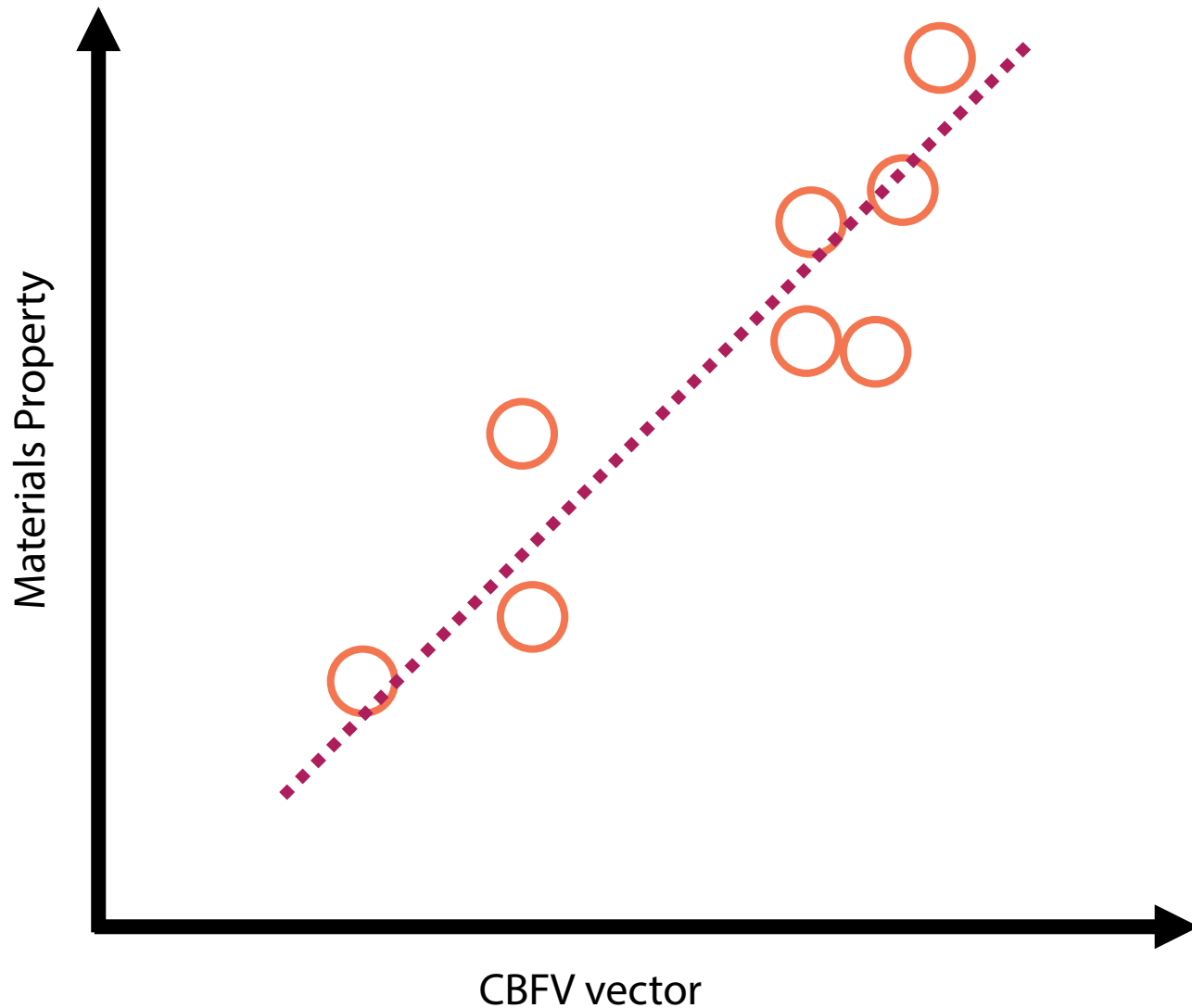
Compounds appear at intermediate positions based on pure elemental constituents

Nearest neighbor is often an OK approximation



Nearest Neighbor:
Find closest datapoint
that already has label

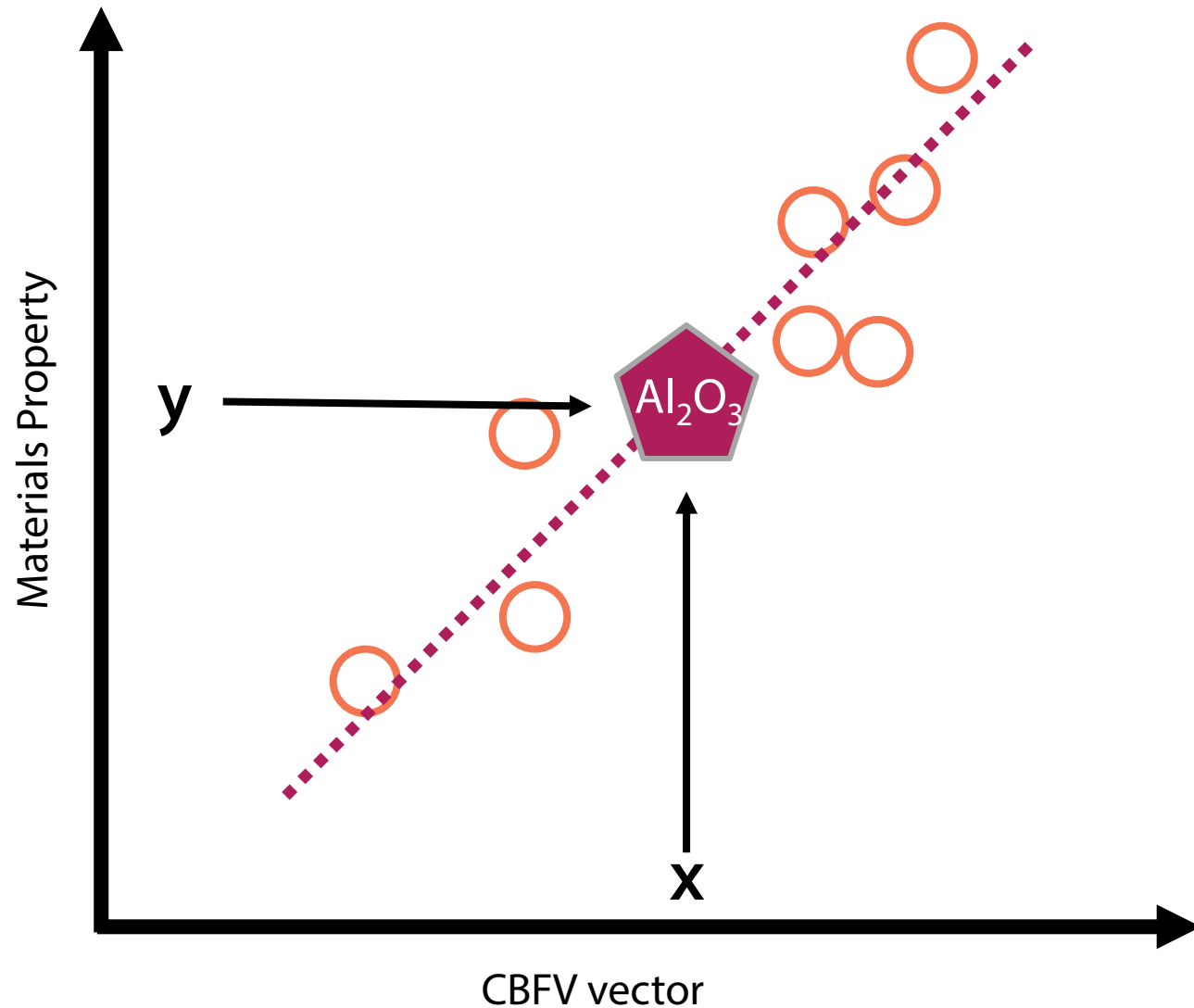
We can fit a trendline to the CBFV vs property



$$y = mx + b$$

Fit line to
known data

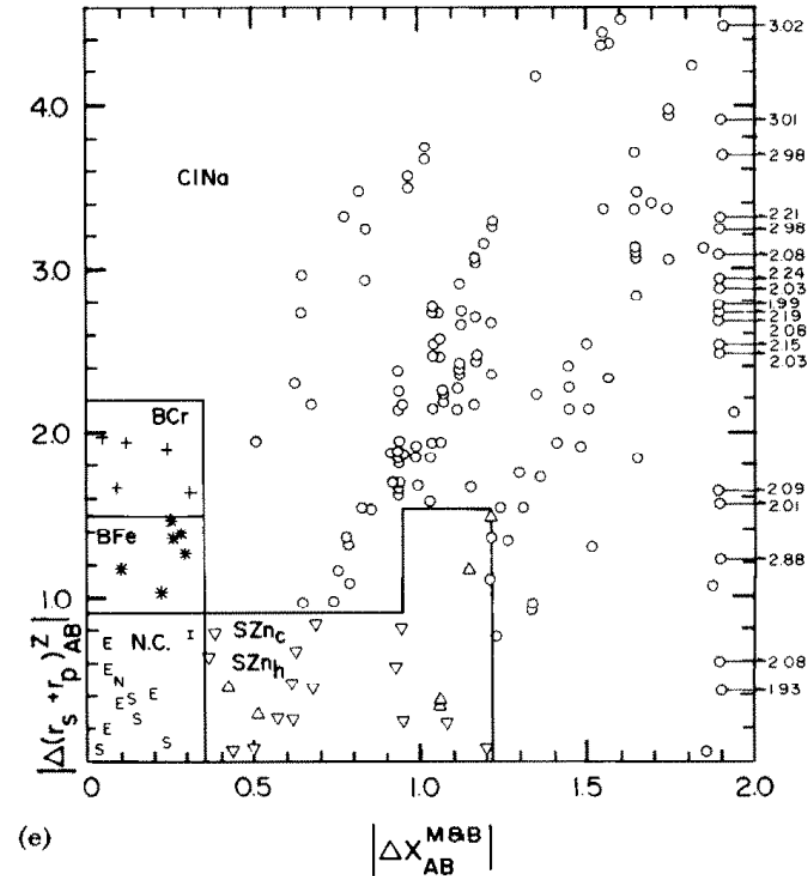
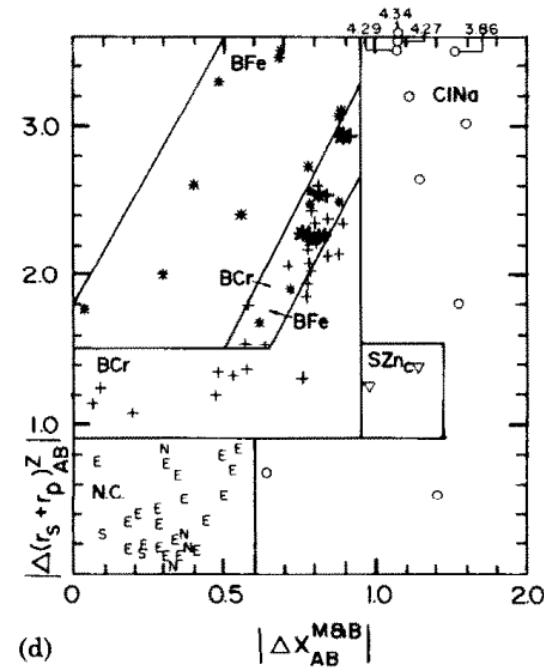
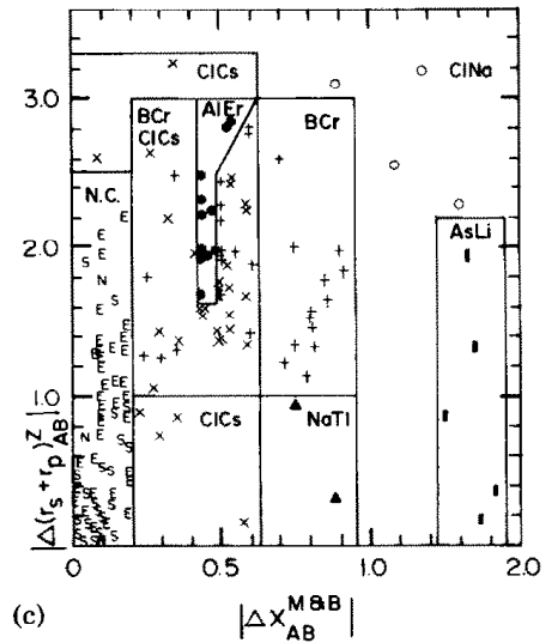
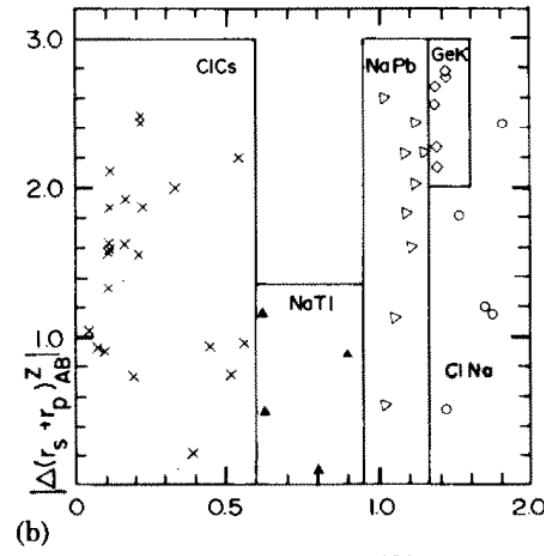
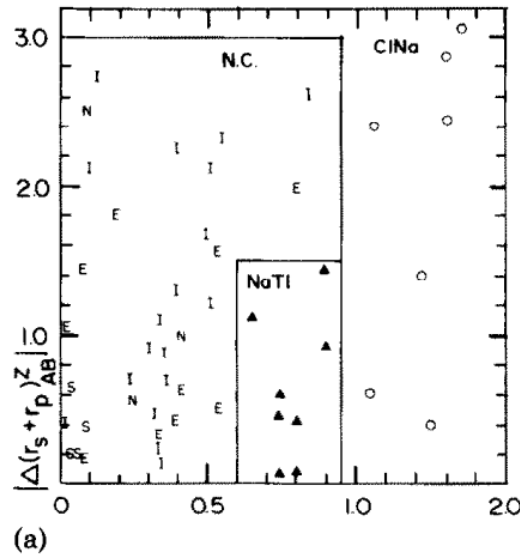
Then, use the trend line to predict properties for new materials



$$y = mx + b$$

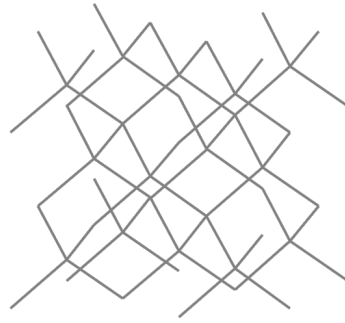
1. Given a new x
2. Generate a prediction y

Villars developed descriptors long before materials informatics!

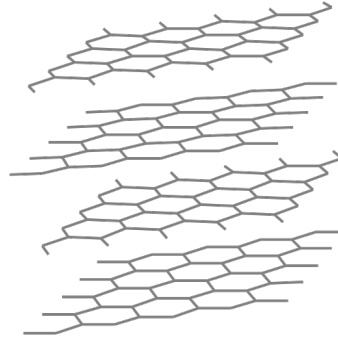


P. Villars, 1983, Journal of Less Common Metals

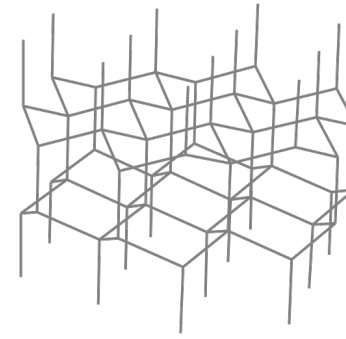
Carbon is a great example of why we can't ignore structure



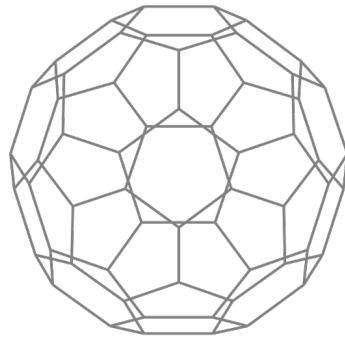
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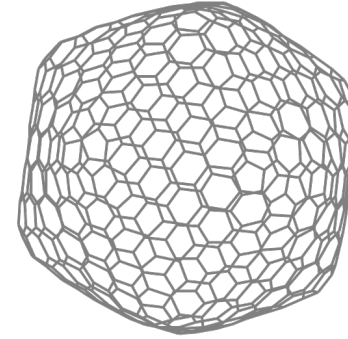
b



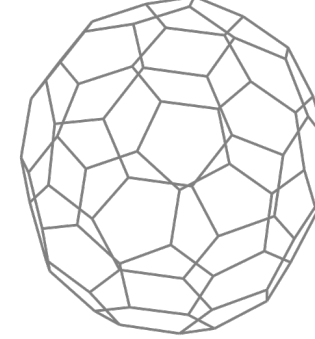
c



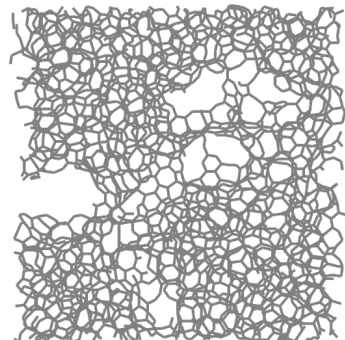
d



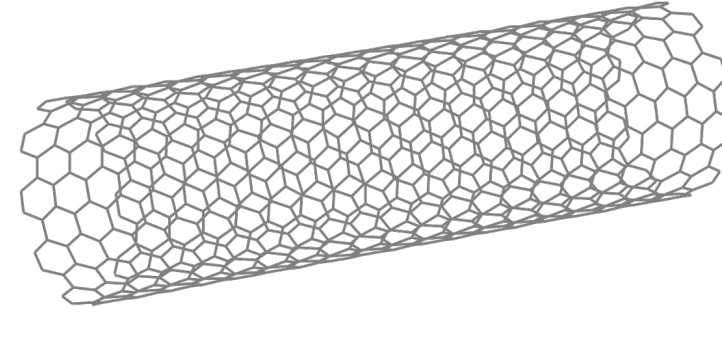
e



f



g



h

Which structure are we talking about for structural feature?

Electronic



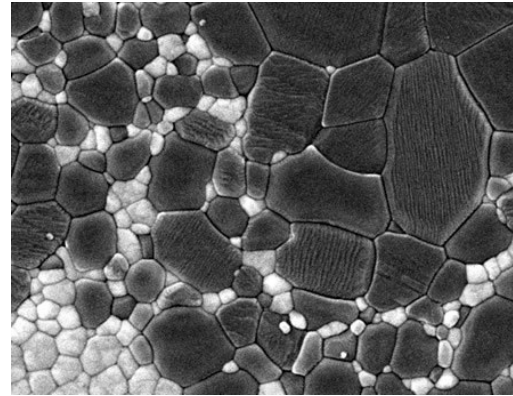
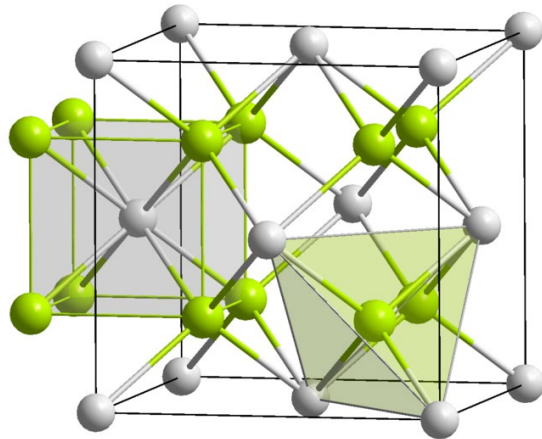
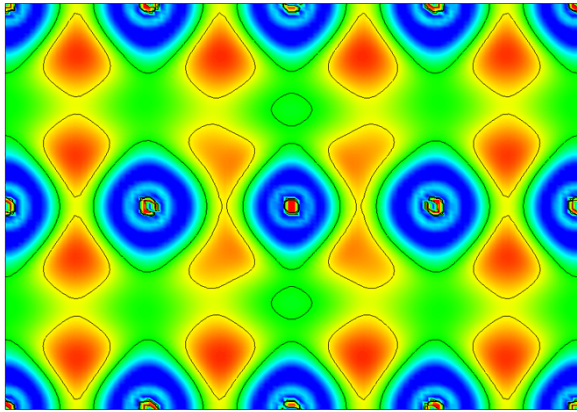
Atomic



Microstructure



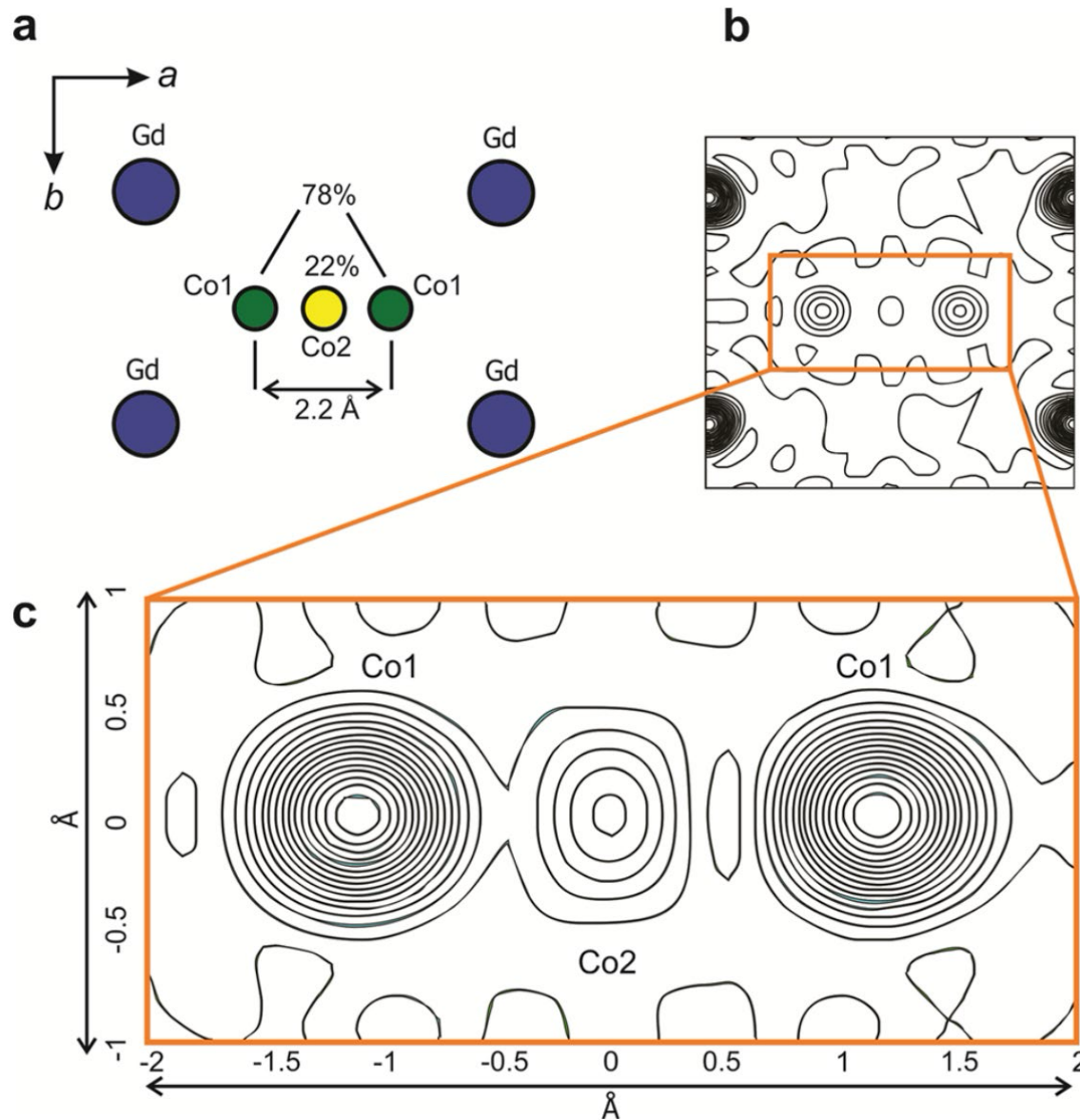
Macroscale



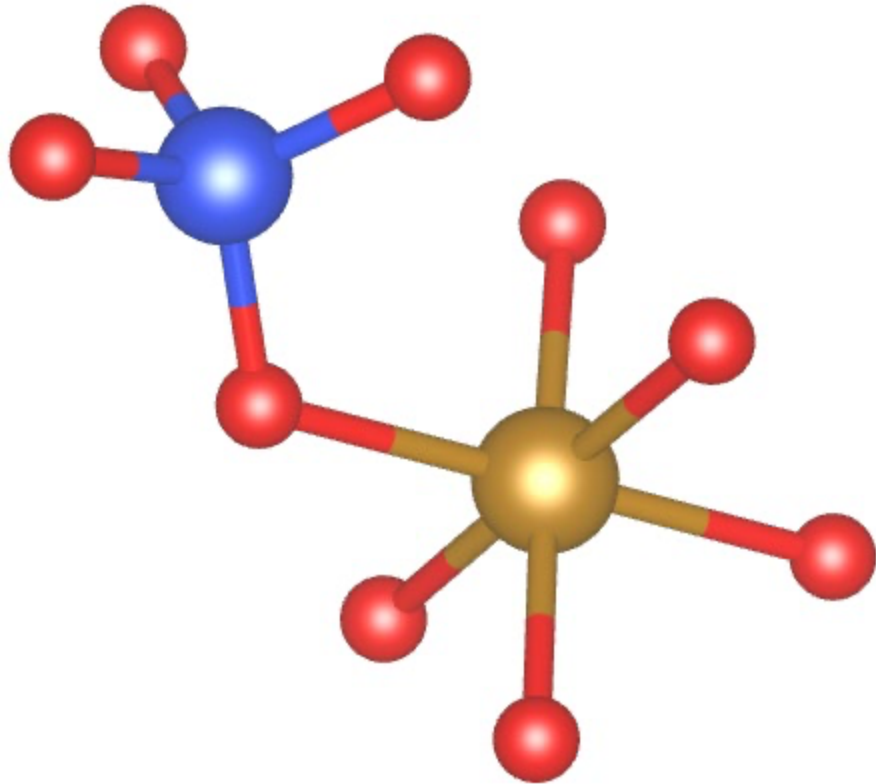
[illegible]

Atomic structure can also be approximated by electron density map

Oliyntyk et al, *Inorg Chem*, 55, 13, 6625-6633, (2016)



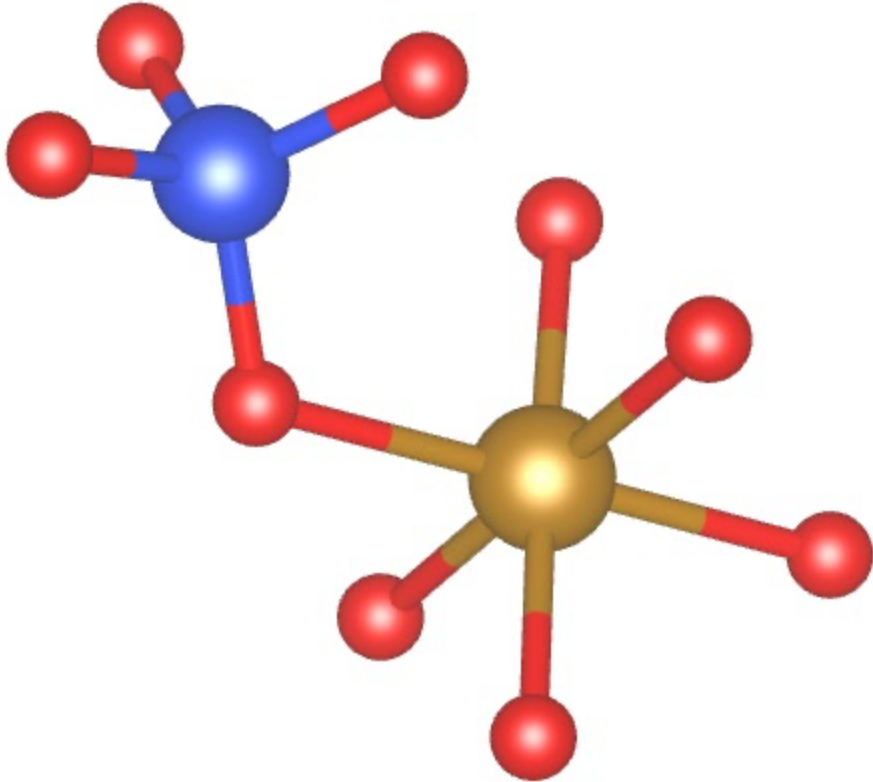
Crystal structure has different features at different scales



Bonds

- Type
- Elements involved
- Length
- Distribution of different bonds

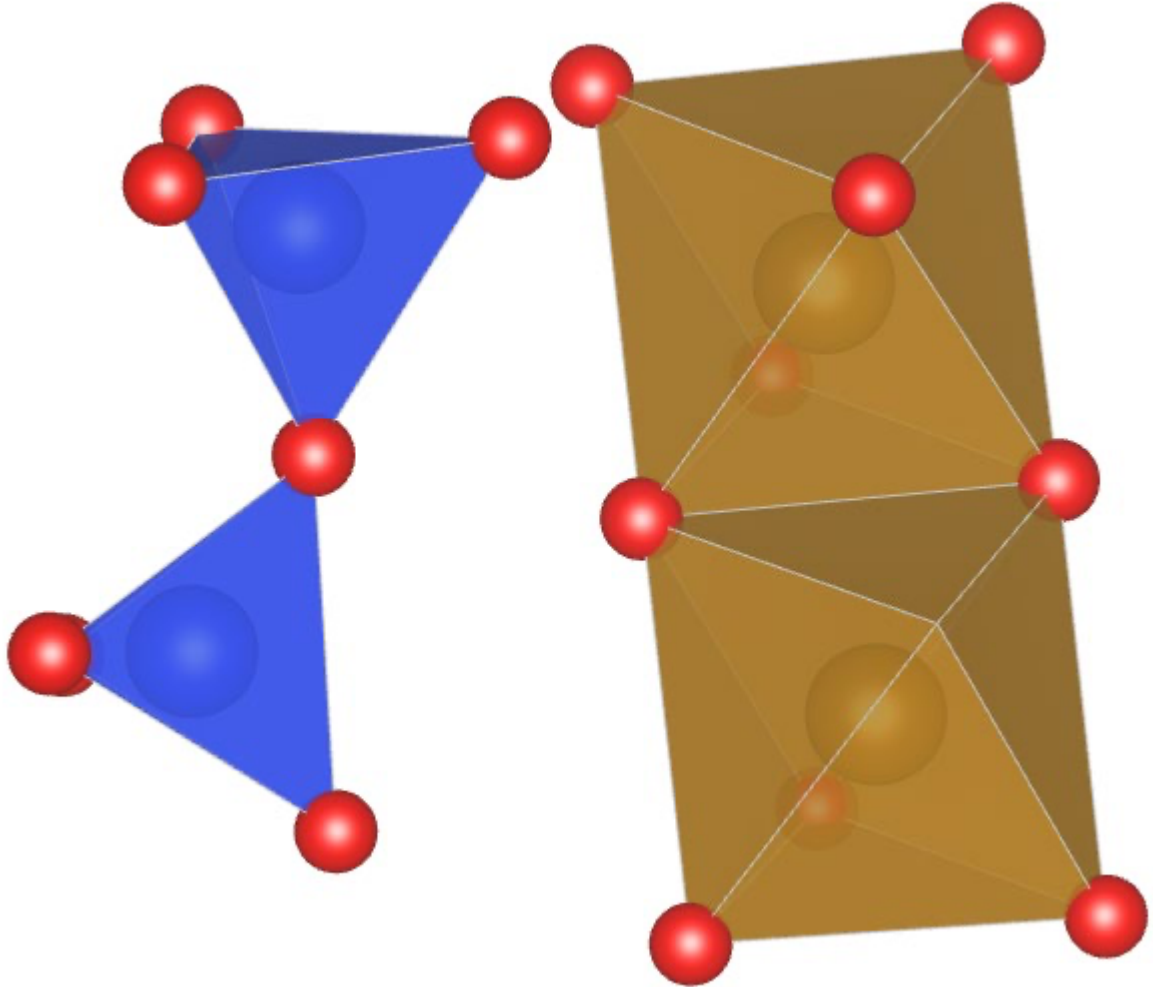
Crystal structure has different features at different scales



Local environment

- Coordination number
- Angles
- Distortions (Jahn-Teller etc)

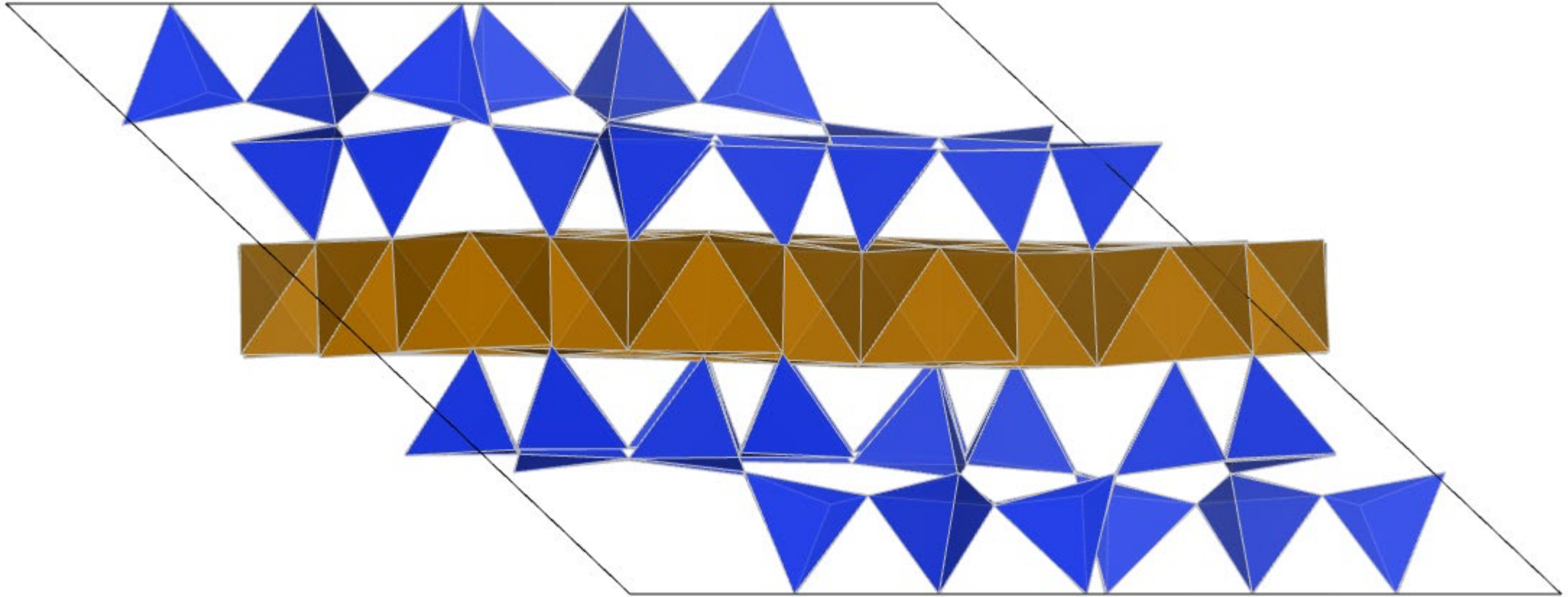
Crystal structure has different features at different scales



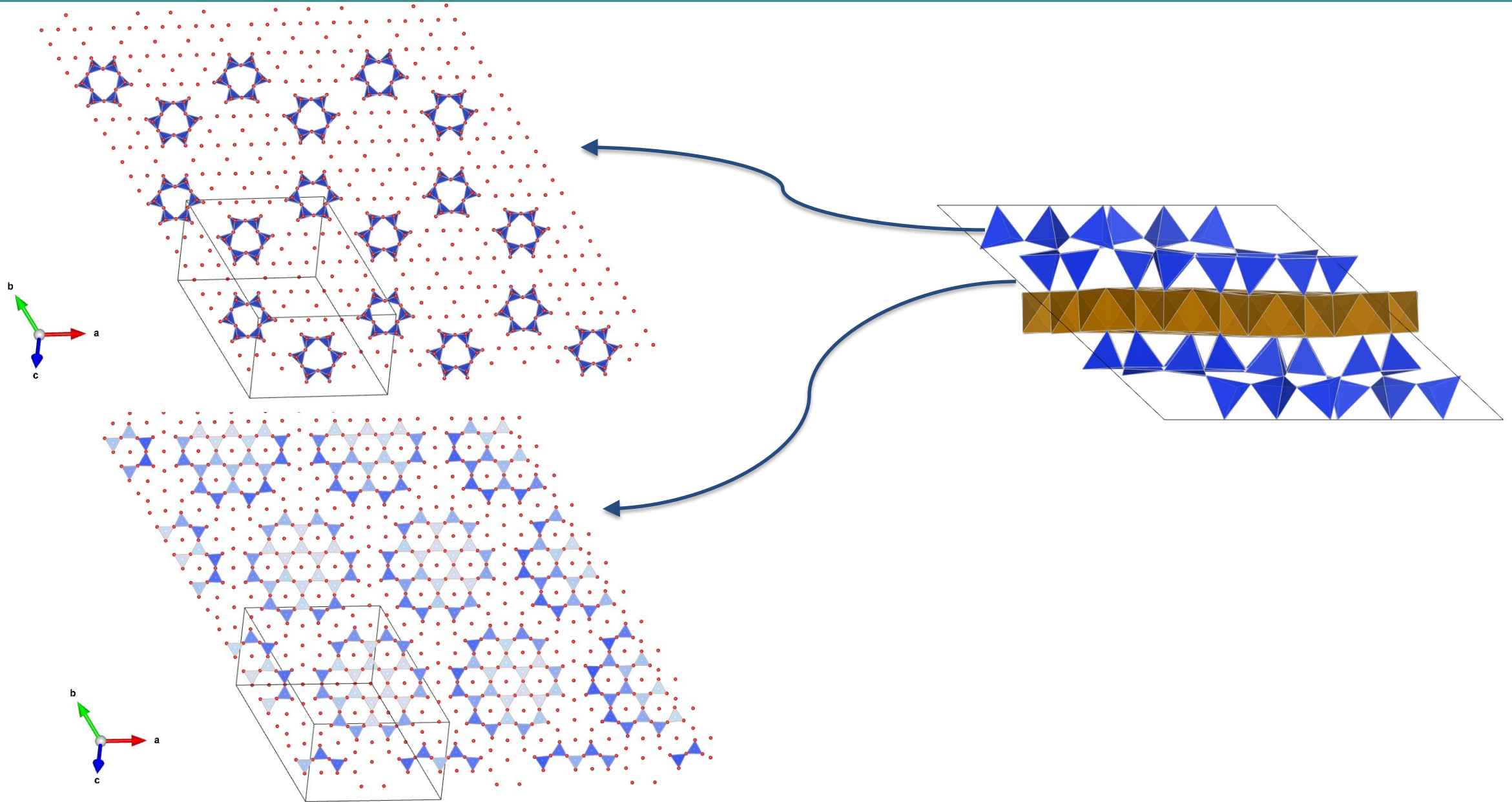
Polyhedra

- isolated vs corner vs edge vs face sharing
- multiple types
- angles
- distances

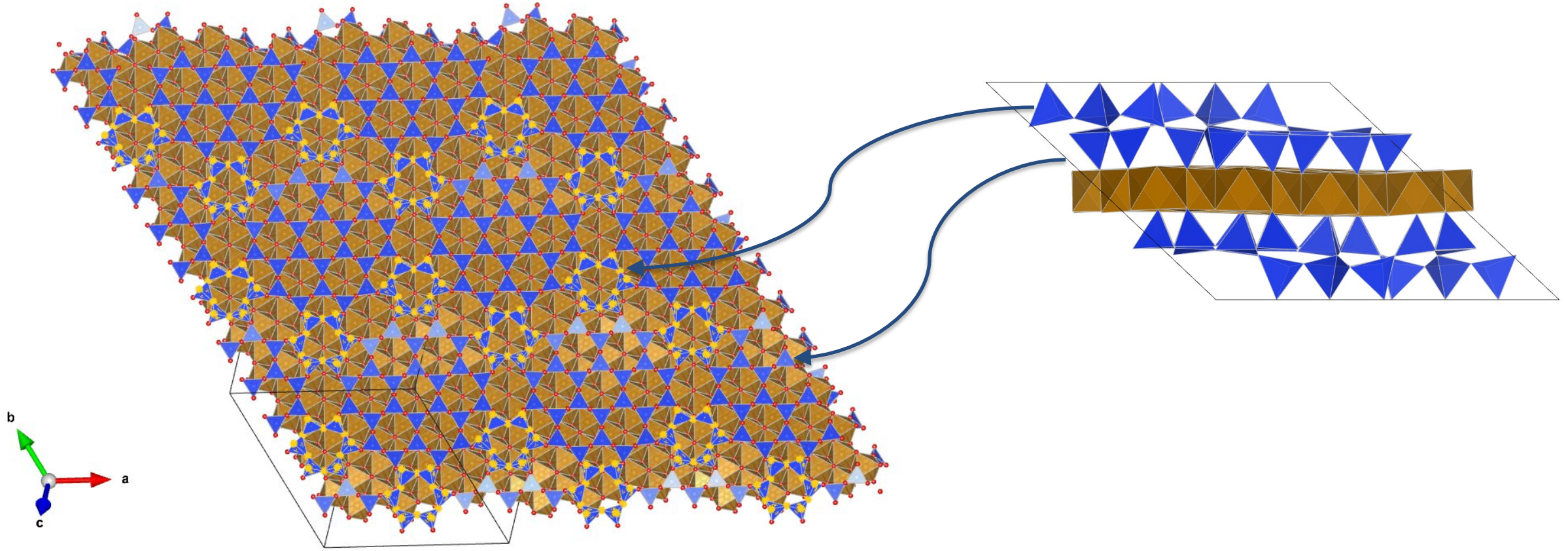
Atomic features can be atomic scale... Or unit cell scale...



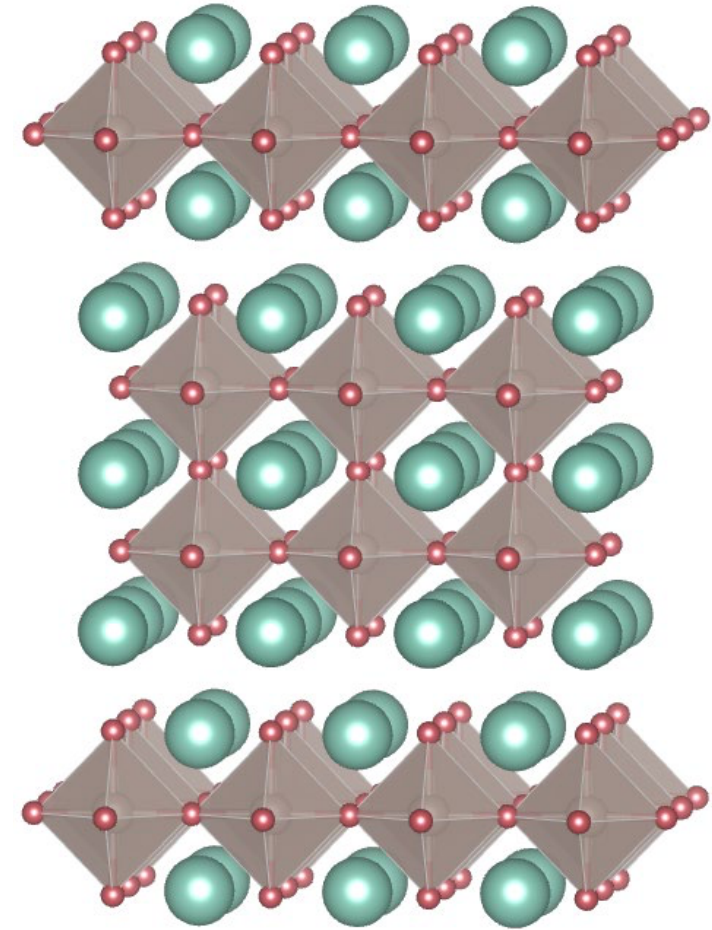
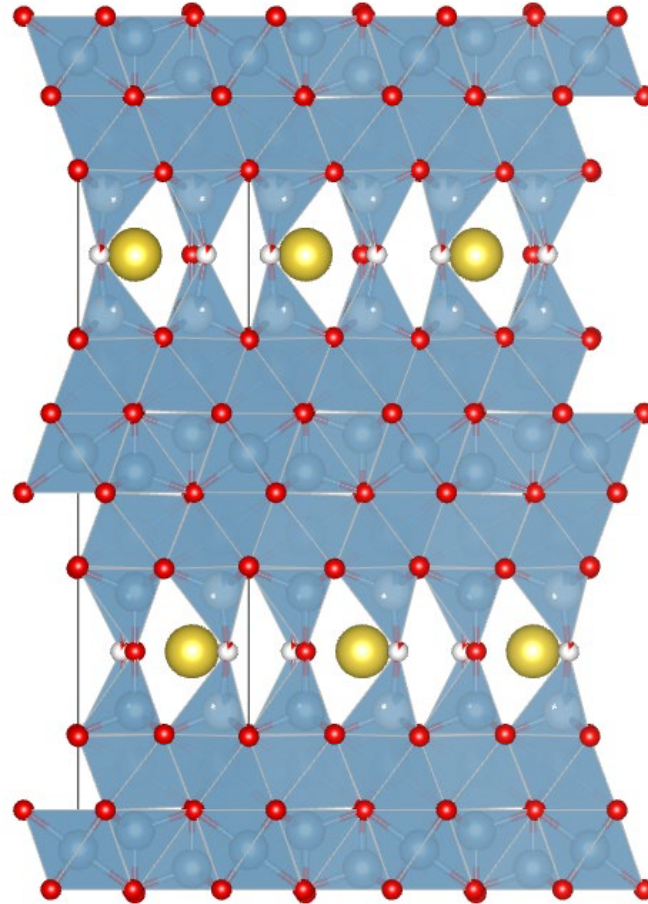
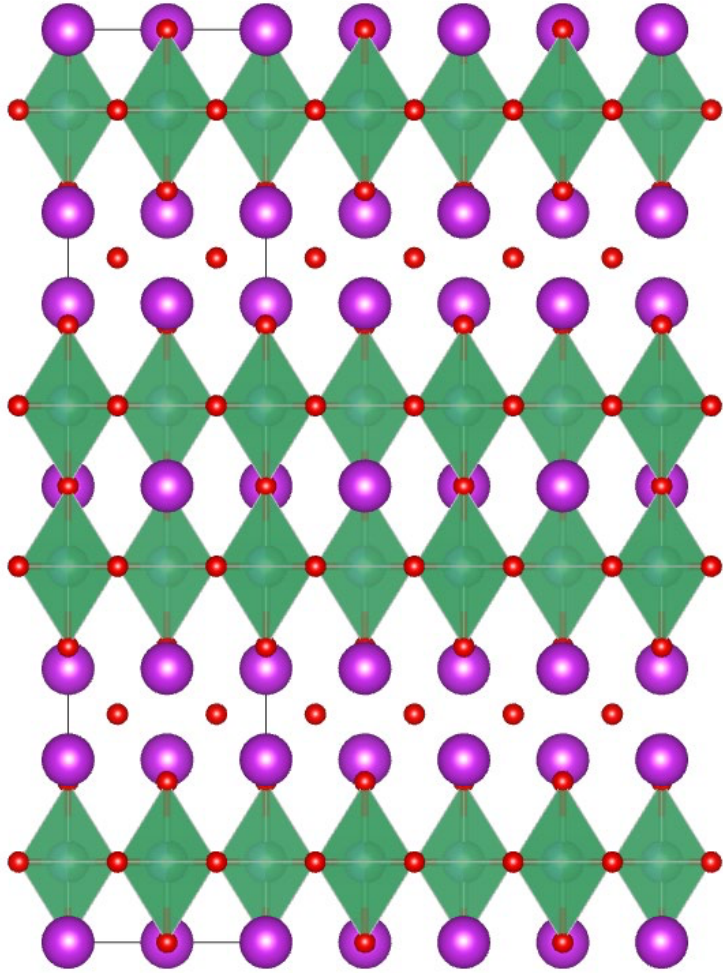
Some unit cell features are “slabs” or layers



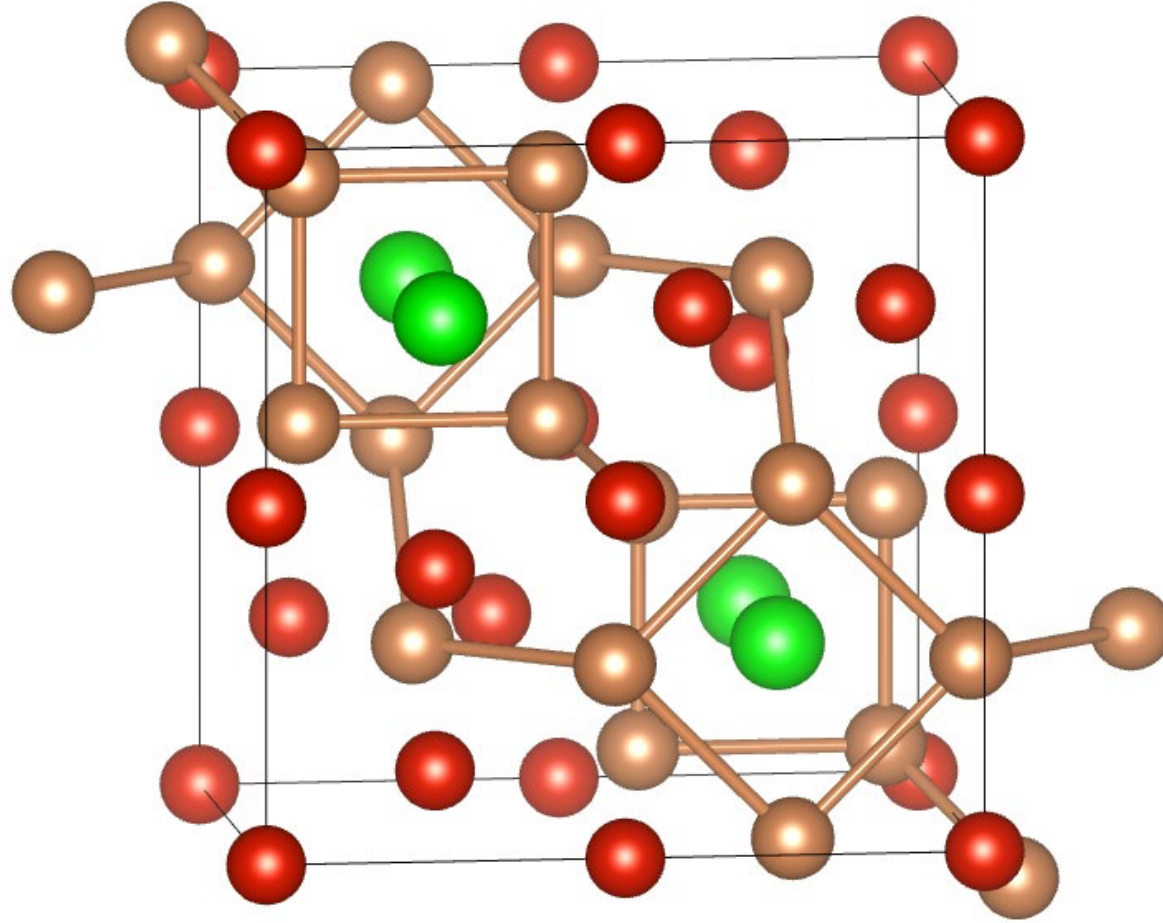
Layers don't need to be continuous



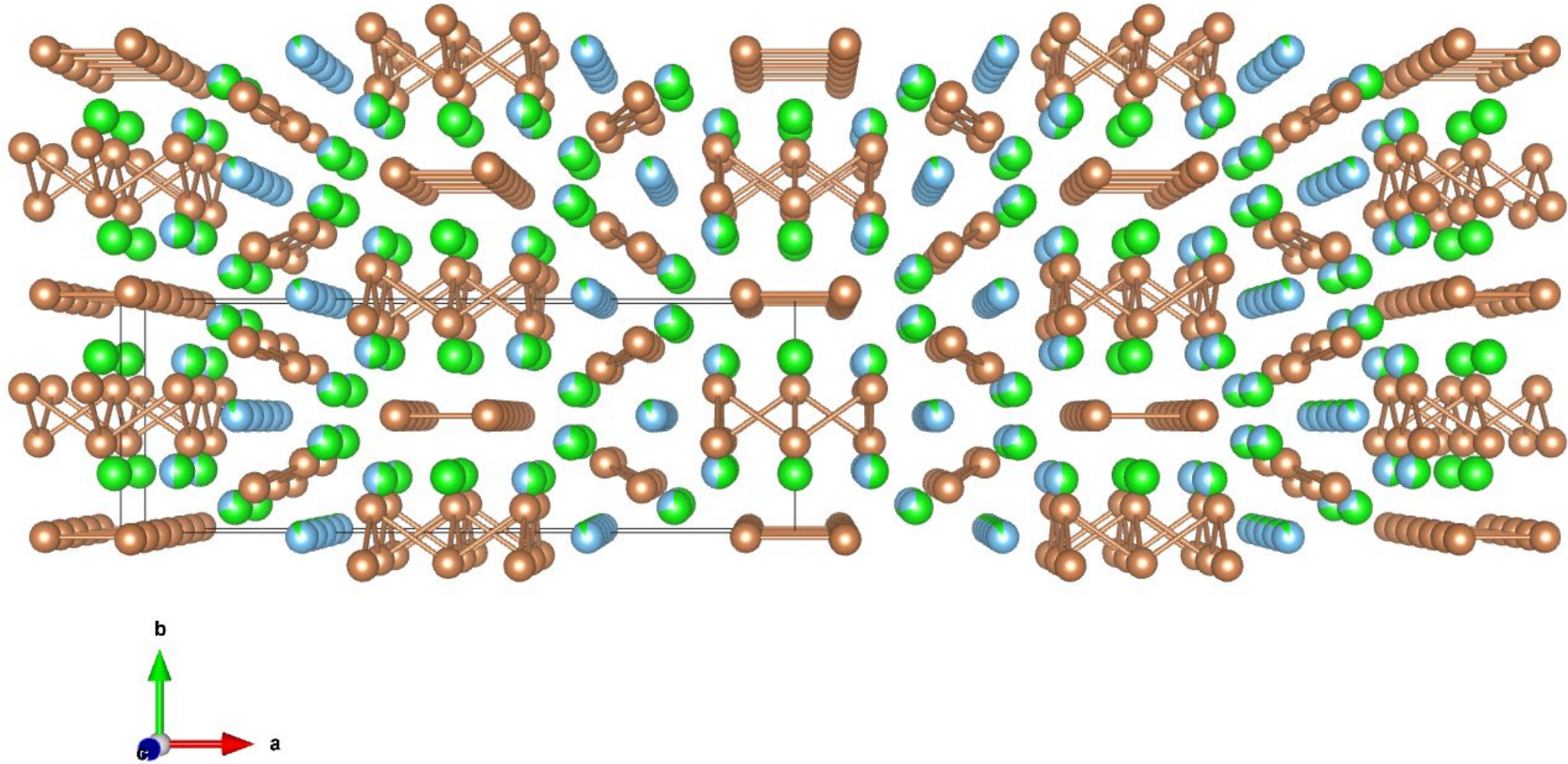
Whole structures are made up of slabs



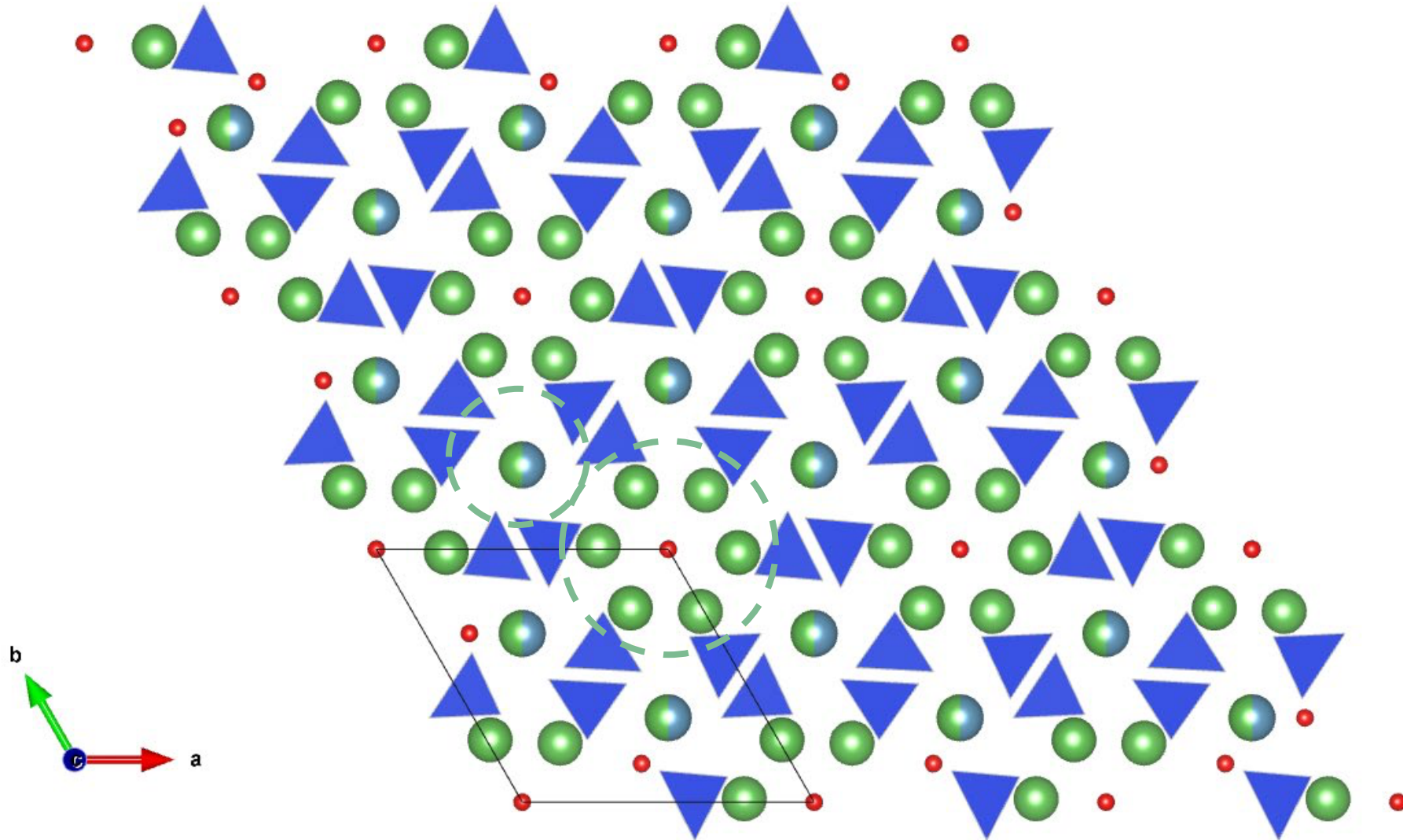
Chains and planes of atoms can define properties



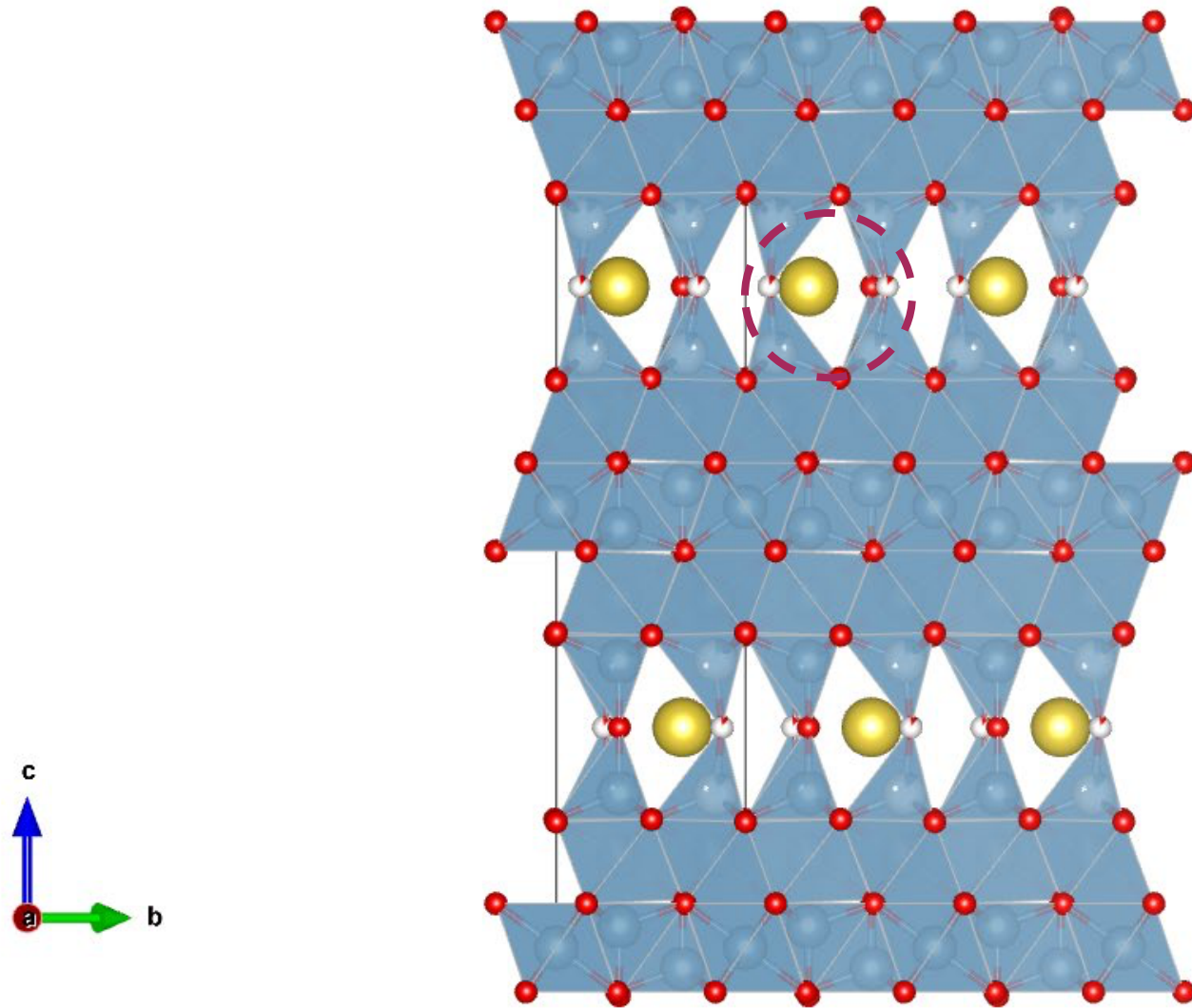
Chains and planes of atoms can define properties



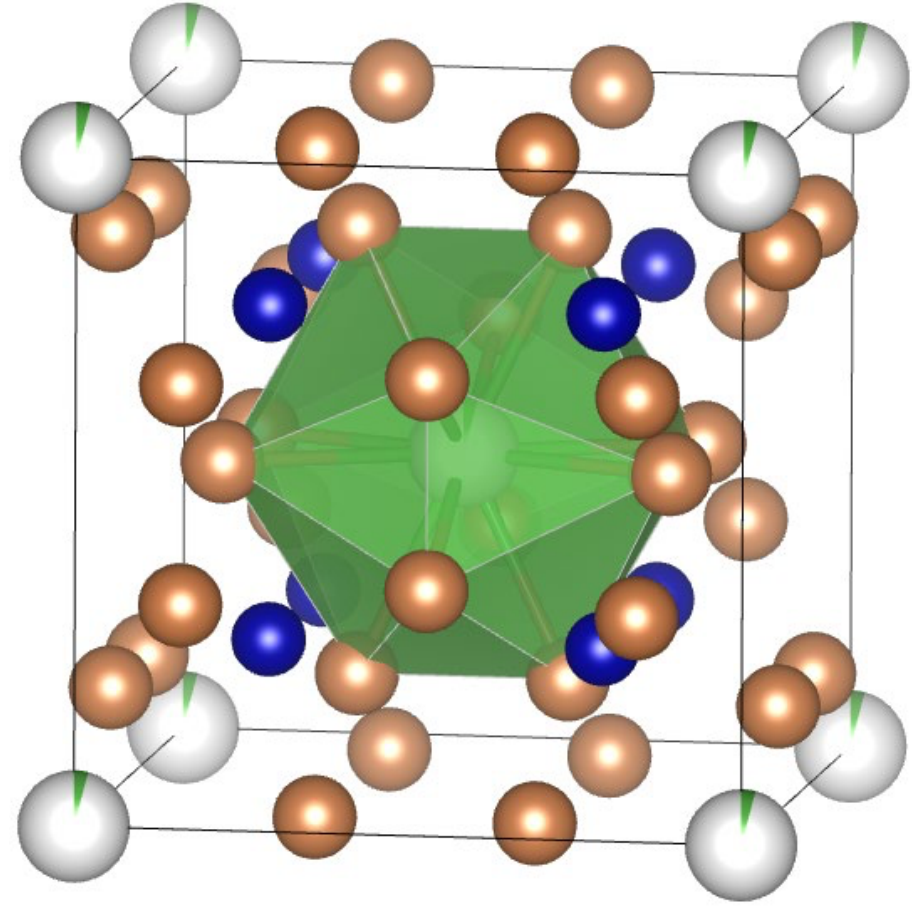
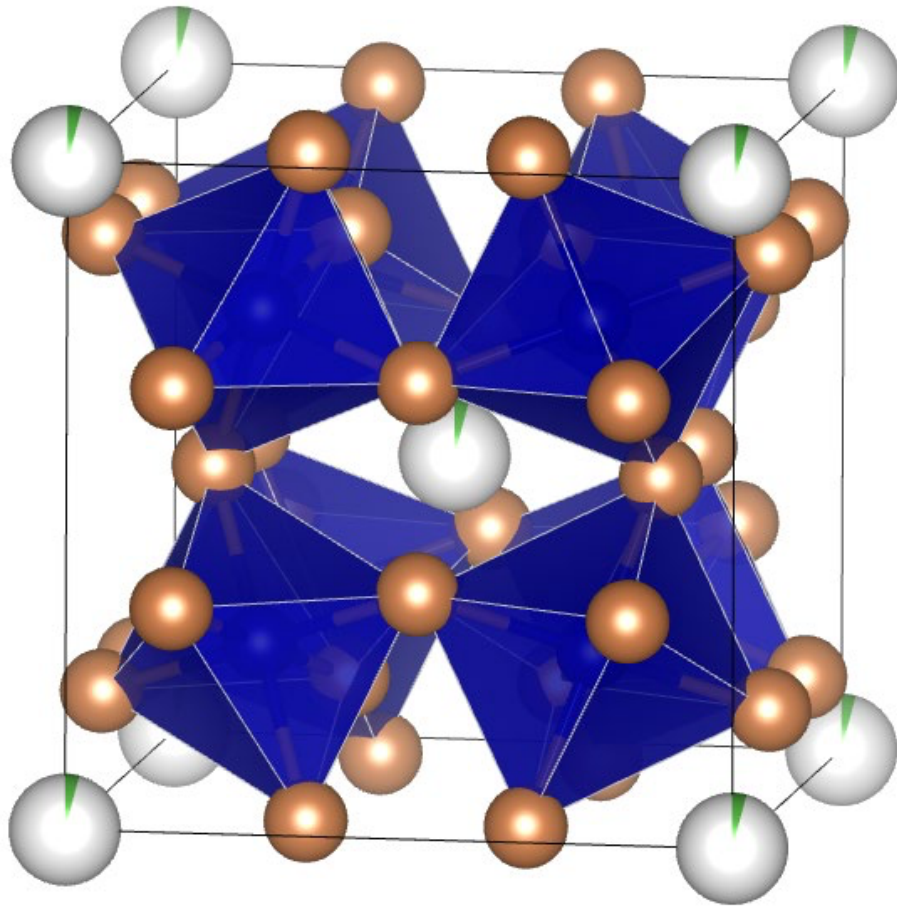
Negative space not occupied by atoms can be the feature



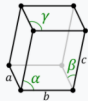
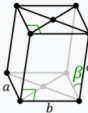
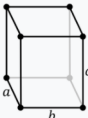
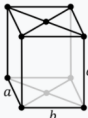
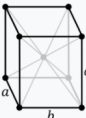
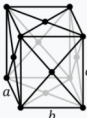
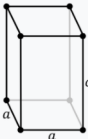
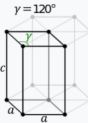
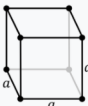
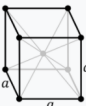
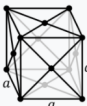
Negative space not occupied by atoms can be the feature



Negative space not occupied by atoms can be the feature



We could define structure with symmetry

Crystal family	Lattice system	Point group (Schönflies notation)	14 Bravais lattices			
			Primitive (P)	Base-centered (S)	Body-centered (I)	Face-centered (F)
Triclinic (a)		C_i				
			aP			
Monoclinic (m)		C_{2h}				
			mP	mS		
Orthorhombic (o)		D_{2h}				
			oP	oS	oI	oF
Tetragonal (t)		D_{4h}				
			tP		tI	
Hexagonal (h)	Rhombohedral	D_{3d}				
	Hexagonal	D_{6h}				
Cubic (c)		O_h				
			cP		cI	cF

7 crystal systems

14 bravais lattices

Lattice parameters (a,b,c)

Angles (α, β, γ)

Centering (primitive, base, body, face)

32 Schoenflies point groups

We could define structure with symmetry

230 The Space Group List Project



230 space groups for unique symmetries
Symmetry is applied to crystal basis

More information at
crystalsymmetry.wordpress.com

Crystal structures drawn with VESTA
K. Momma and F. Izumi, *J. Appl. Crystallogr.* **2011**, 44, 1272–1276

How do we encode all of these features?

Structural feature	Specific structure
Layered	Yes
Channels	No
RS blocks	Yes
Octahedra	Yes
Tetrahedra	No
...	
Corner-sharing	Yes
Edge-sharing	No

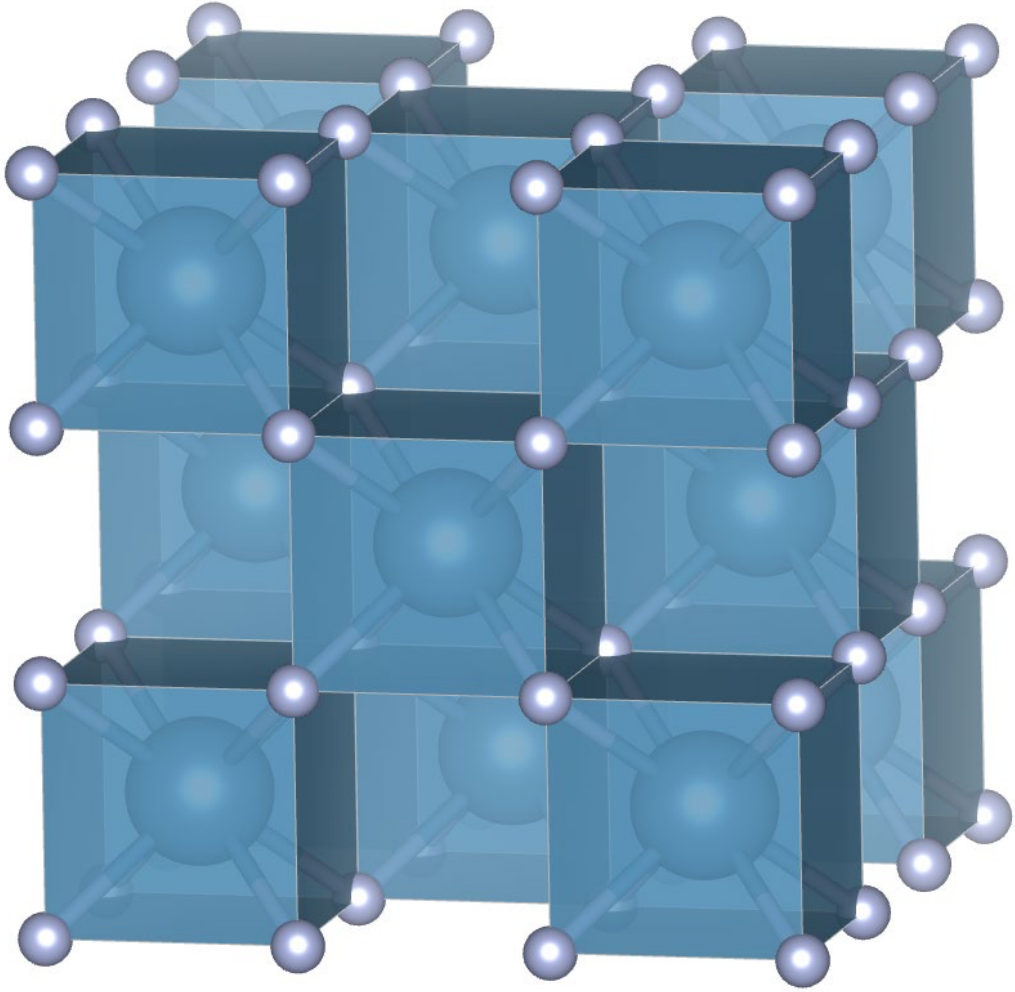
Maybe we could one-hot encode them?

How do we encode all of these features?

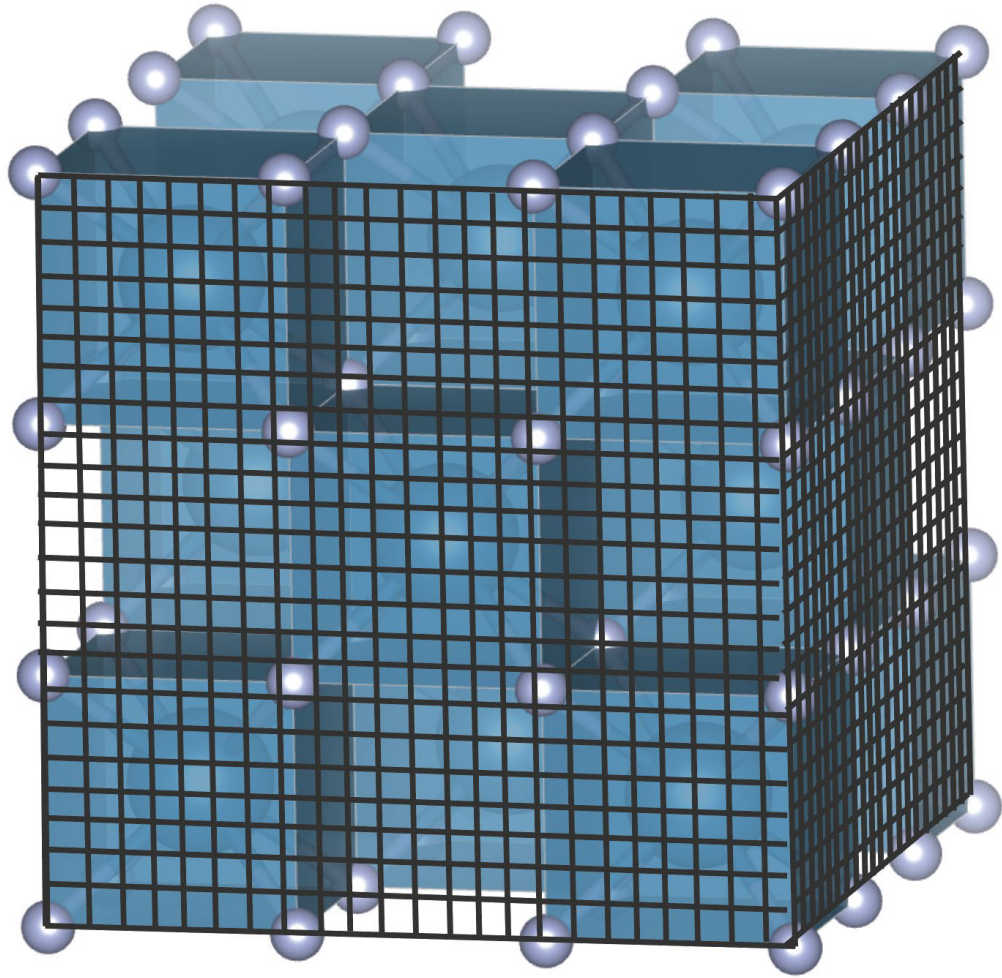
Structural feature	Ruddlesden-Popper	Aurivillius	Spinel	Rock Salt
Layered	1	1	0	0
Channels	0	0	1	0
RS blocks	1	0	0	1
Octahedra	1	1	1	1
Tetrahedra	0	0	1	0
...				
Corner-sharing	1	1	1	1
Edge-sharing	0	0	1	0

Maybe we could one-hot encode them?

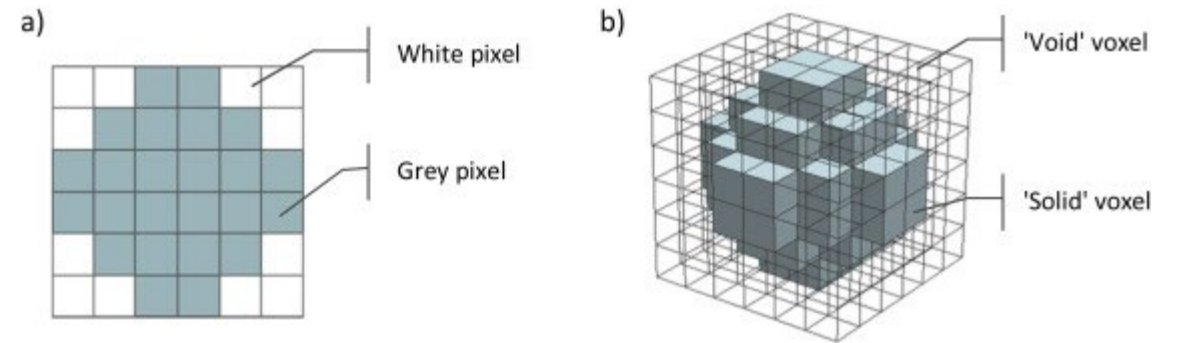
We could chop the crystal up into tiny voxels occupied or not



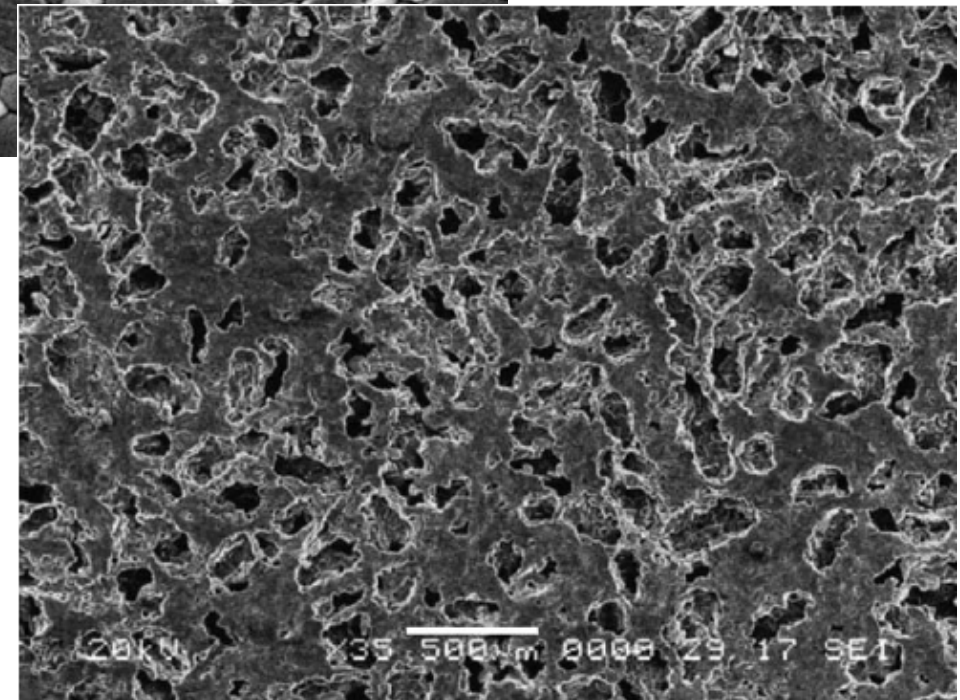
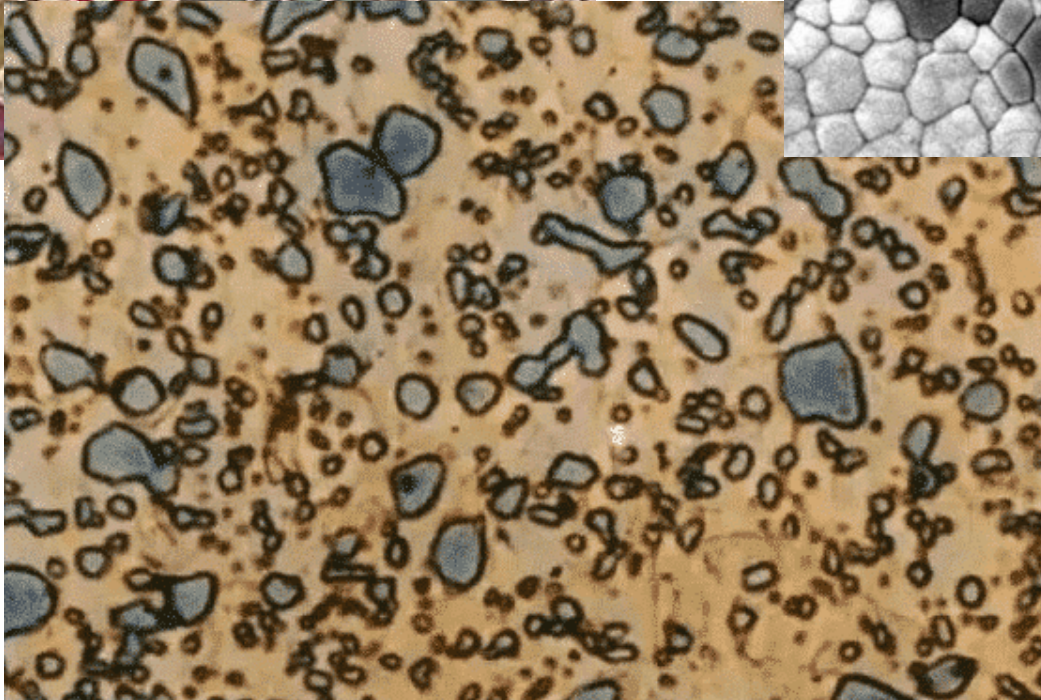
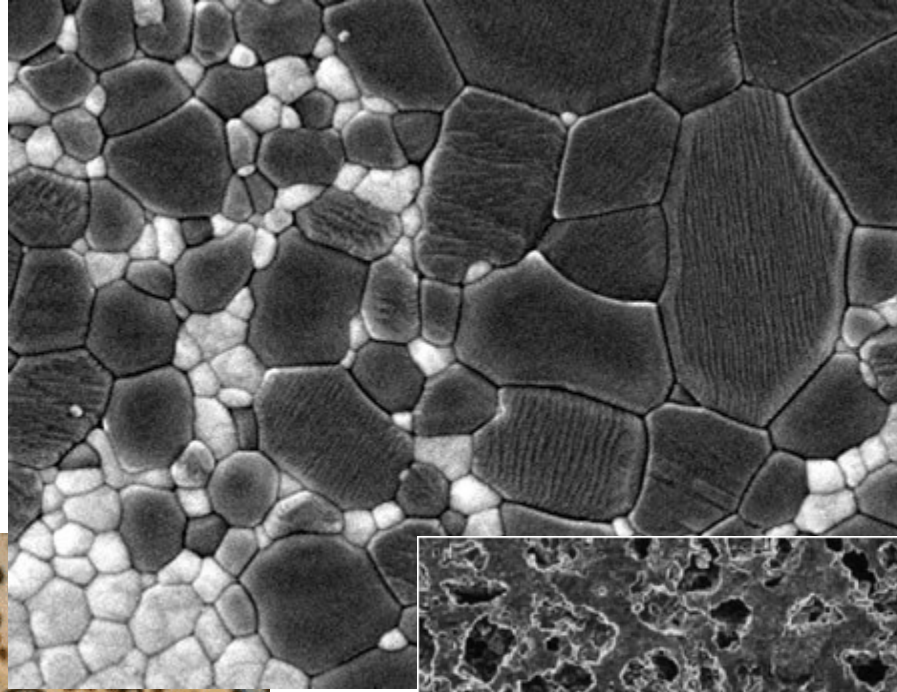
We could chop the crystal up into tiny voxels occupied or not



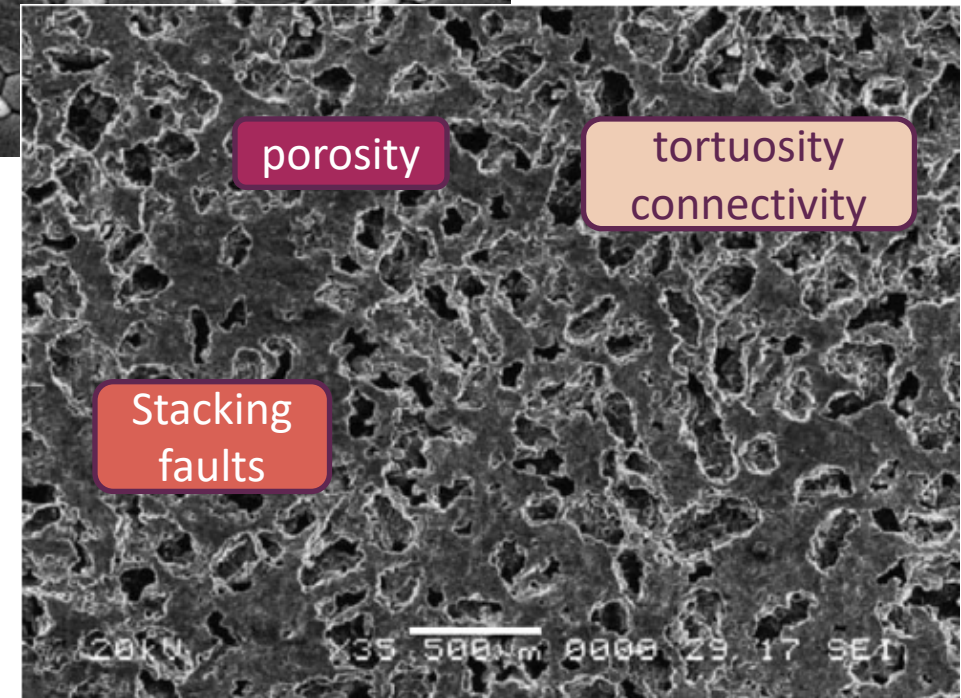
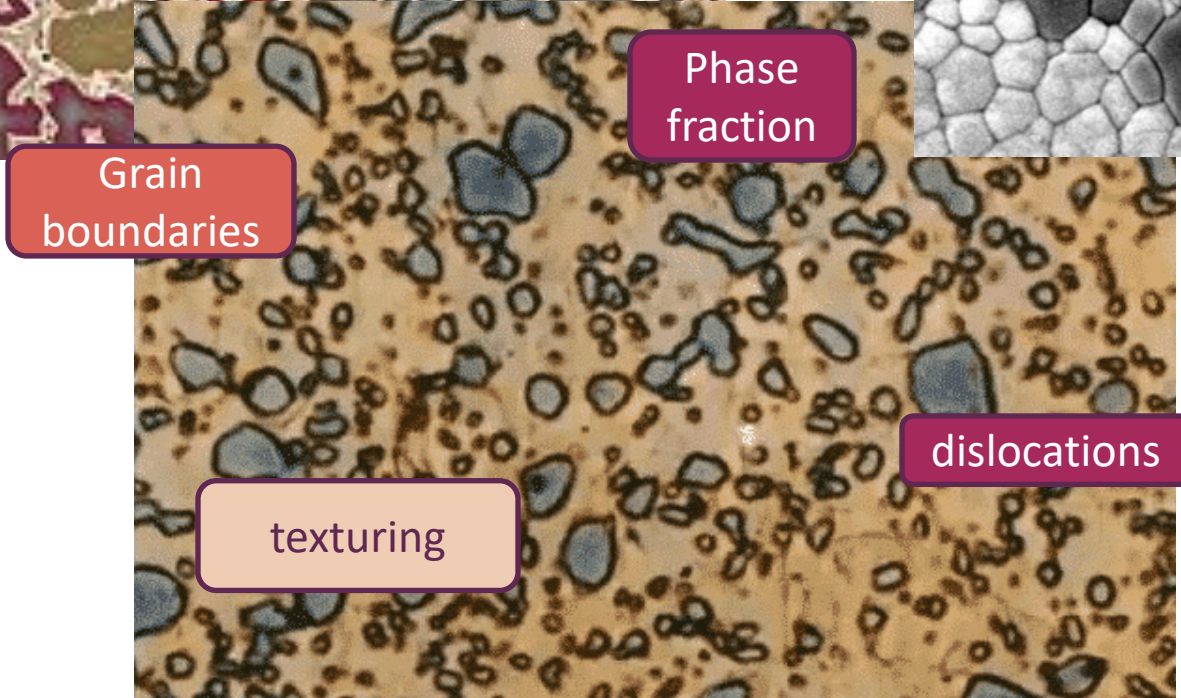
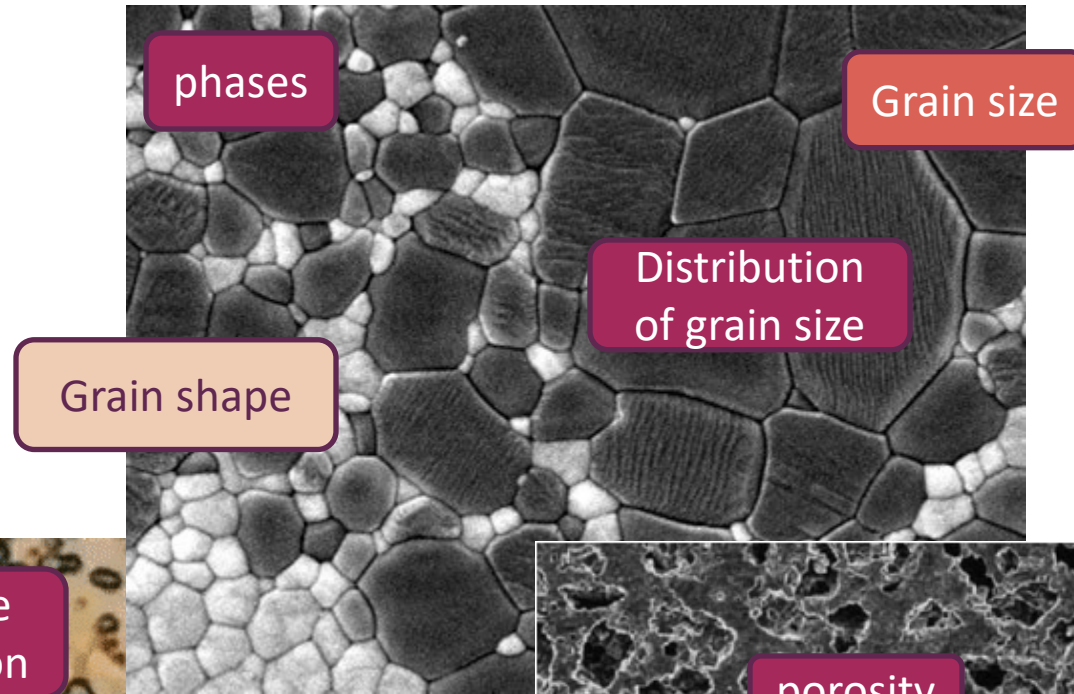
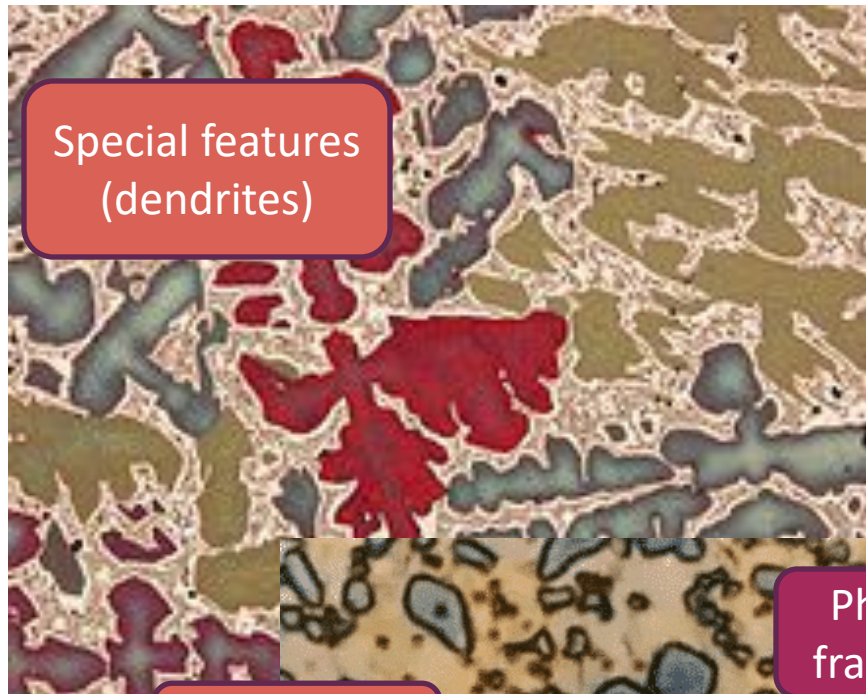
Do put a point at atom center, or do we try to account for atom size?
How do we encode different atom chemistries?



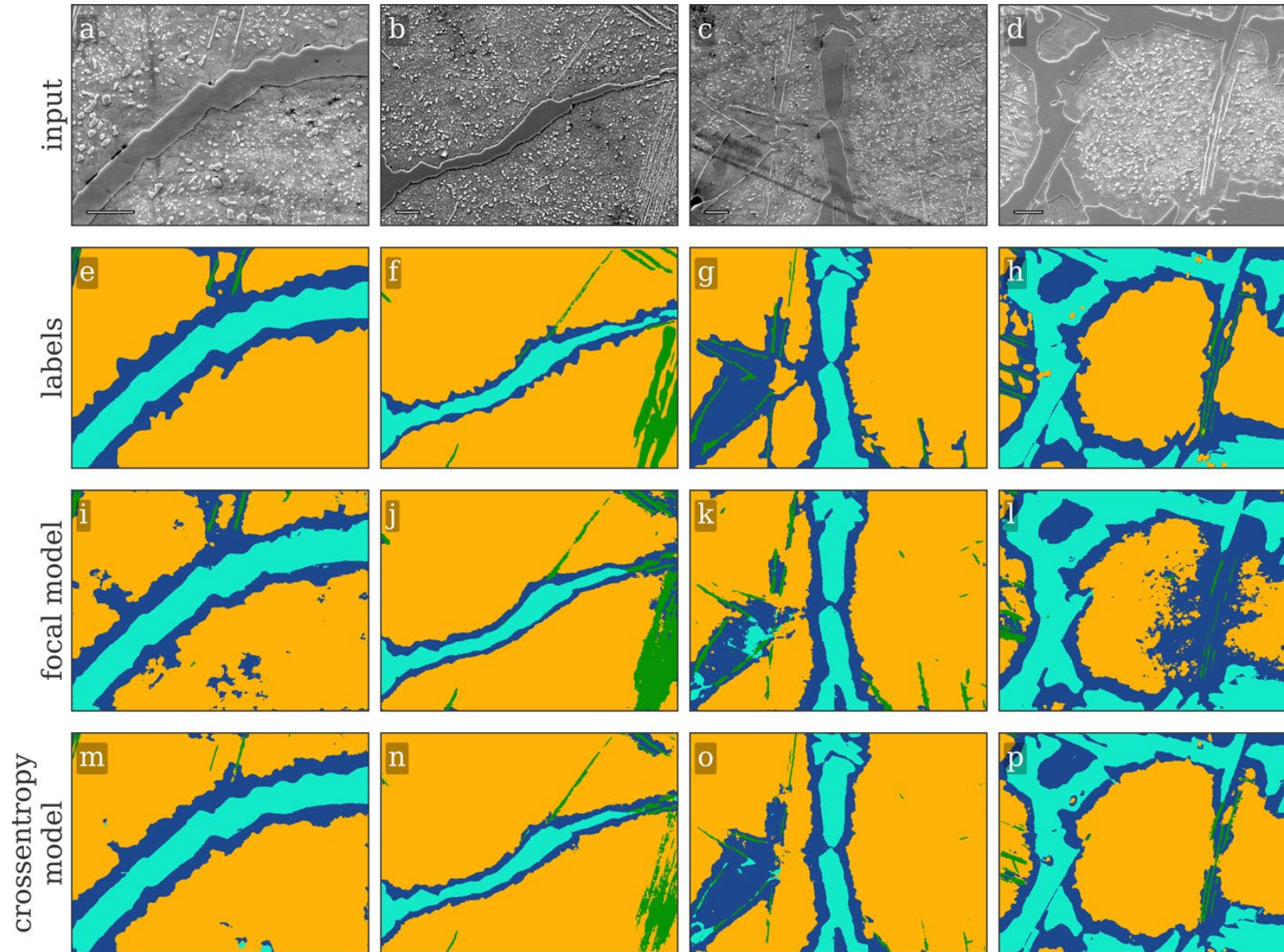
What information is present at the microstructure scale?



What information is present at the microstructure scale?

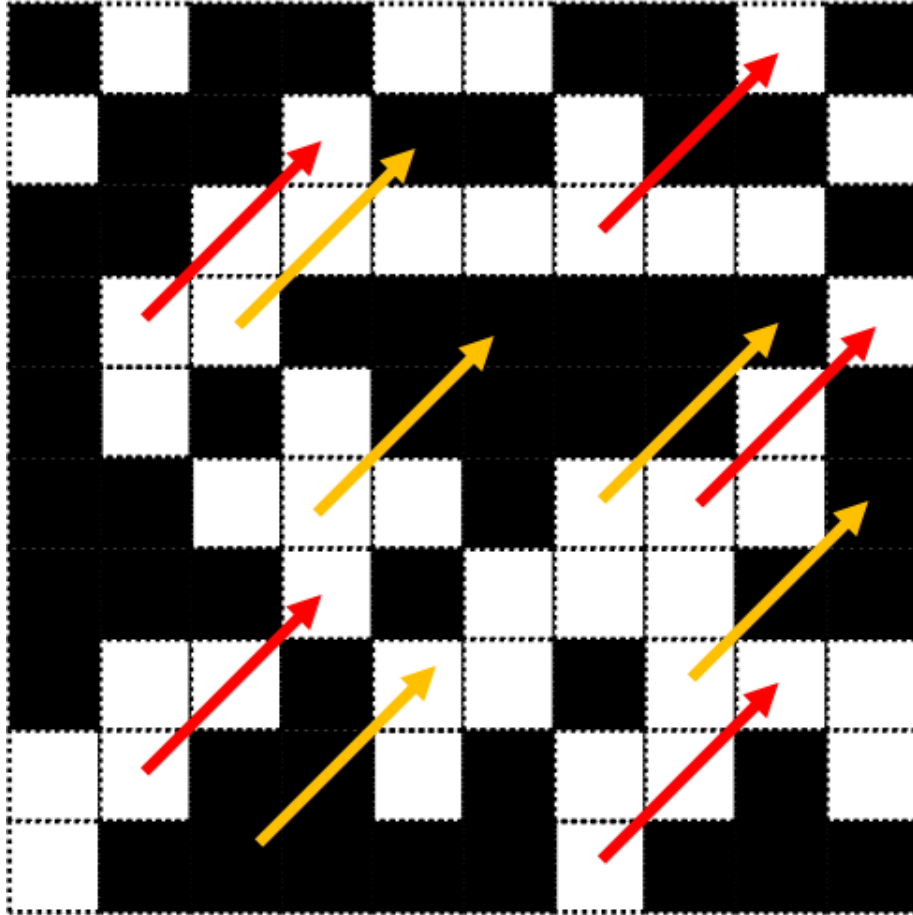


How do we identify and featurize these features?

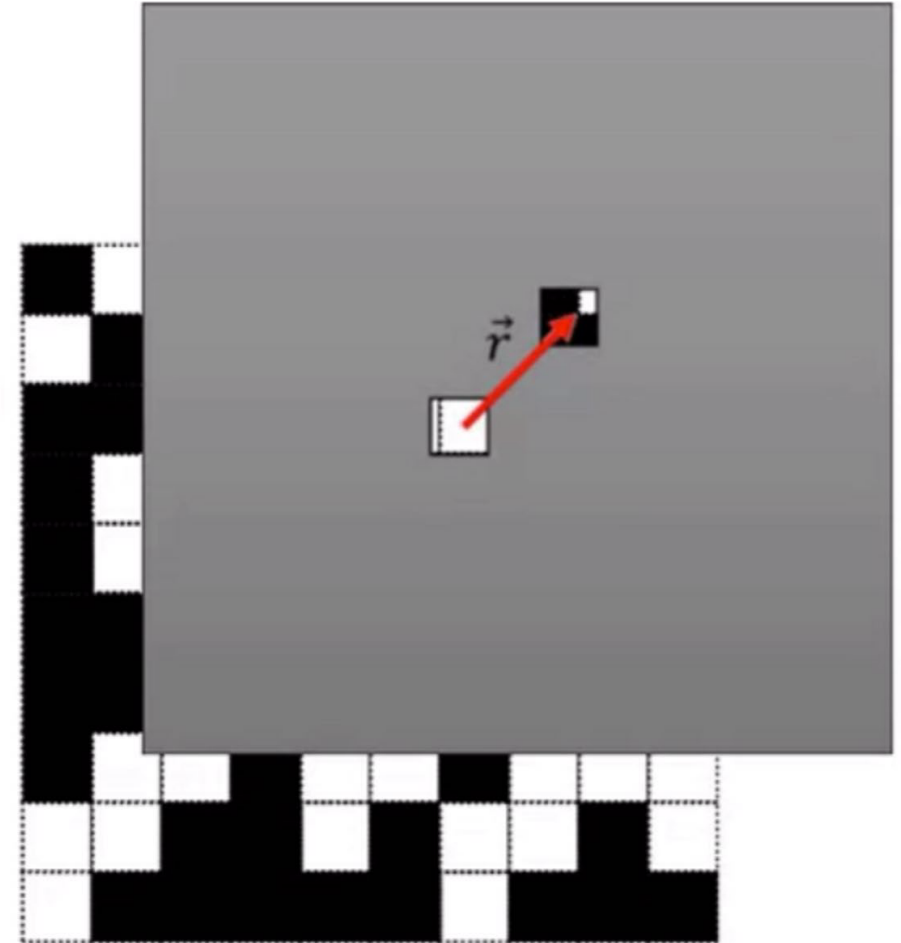


2-pt statistics is one powerful tool for quantifying these features

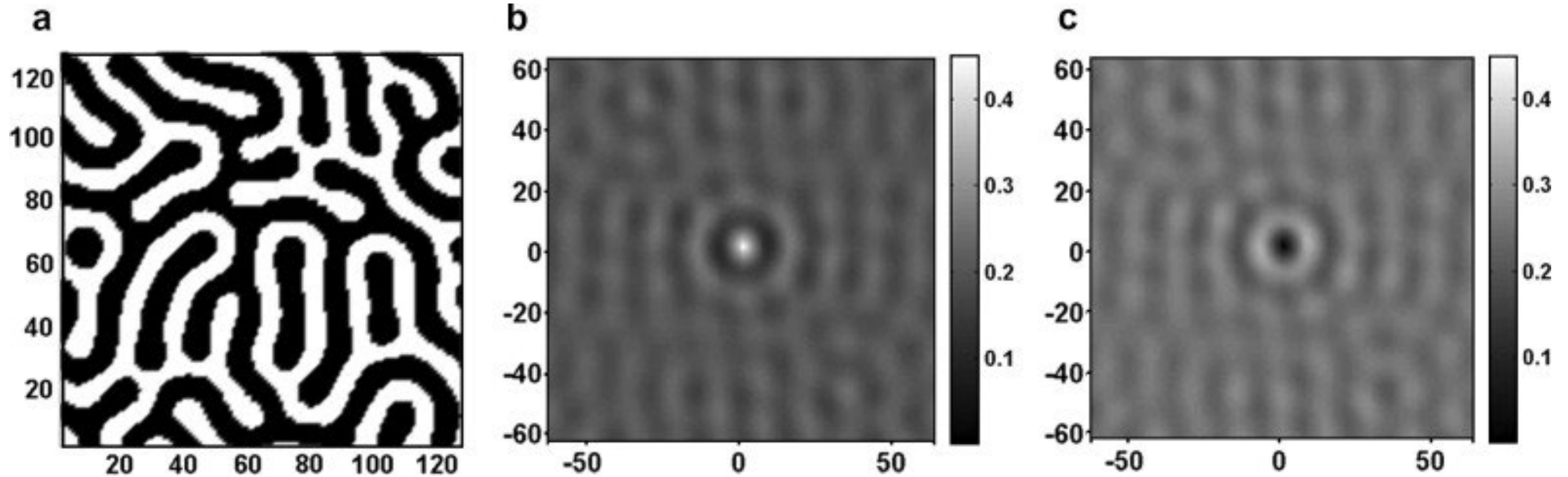
$$f_r^{hh'} = \frac{\# \text{ Trials Successful}}{\# \text{ Trials Attempted}}$$



$$P(A^h \cap A_{@ \vec{r}}^{h'} | \vec{r}) = \frac{P(A^h \cap A_{@ \vec{r}}^{h'} \cap \vec{r})}{P(\vec{r})}$$



2-pt statistics is one powerful tool for quantifying these features

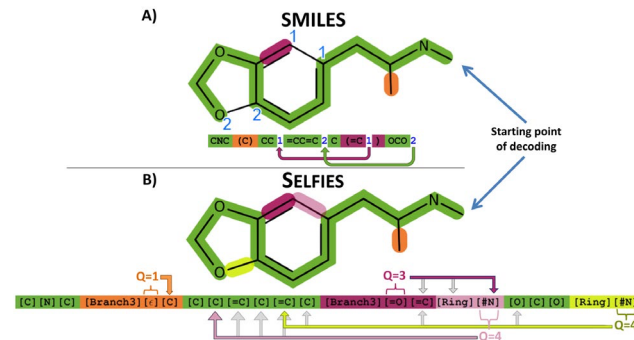


Fullwood, David T., Stephen R. Niezgoda, and Surya R. Kalidindi. *Acta Materialia* 56.5 (2008): 942-948.

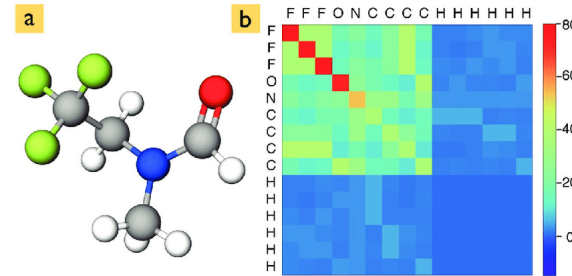
Microstructure
segmentation/analysis



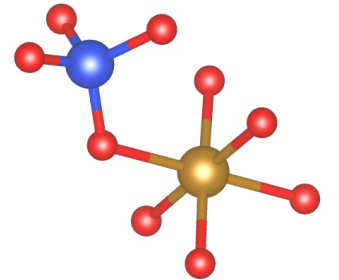
String representations of molecules



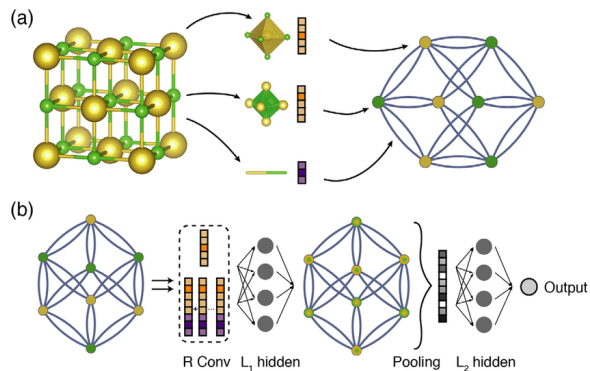
Coulomb matrix



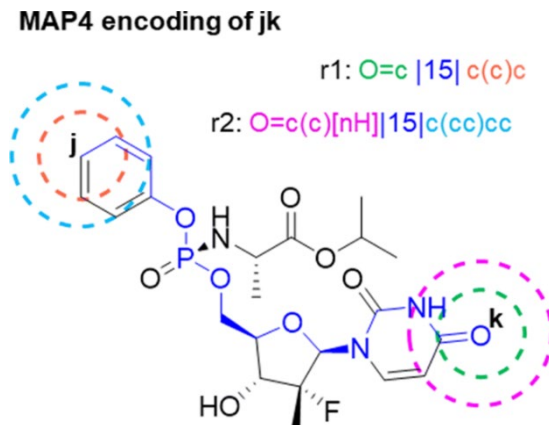
3D coordinates



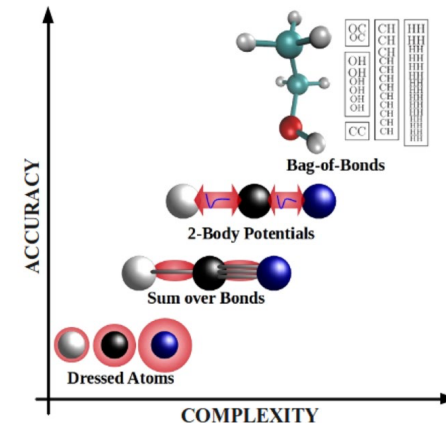
Crystal graphs



Fingerprints



Bag of Bonds /
Fragments



Electron density

